

Recommendations

Table 1. Key Recommendations		
Recommendations for BSP	Priority	Timeframe 1/
Resources/ Project Management		
Assign BSP staff with diverse, multi-disciplinary expertise to the dedicated working teams for each identified wCBDC use case (paragraph 299).	High	Near term
Identify overlapping action items from the identified use cases in the roadmap and assess opportunities to optimize efforts and streamline process across both wCBDC use case workstreams (paragraph 3030).	Medium	Near term
Examine the dependencies of the action items for each use case and allocate timeline for their completion (paragraph 3131).	Medium	Medium term
Communication		
Continue to engage and communicate with stakeholders to clarify the value propositions of wCBDC's identified use cases and gain their buy-in for the roadmap implementation (paragraph 32).	High	Medium term
Develop a joint communication strategy and outlets with key partners to ensure consistent messaging throughout various stages for implementation of each use case (paragraph 33).	Medium	Medium term
Promote public awareness and education to support digital and financial literacy, to the extent they relate to wCBDC's policy goals (paragraph 34).	Medium	Medium term
Success Measurement and Continuation of the Roadmap		
Continue working within the IMF's 5P methodology and establish key performance indicators (KPIs), go/no go decision points, and success metrics for each wCBDC use case to measure progress and effectiveness (paragraph 35).	High	Medium term
Continue to refine the draft of the roadmap and engage with other internal departments and external stakeholders for alignment (paragraph 36).	Medium	Near term
Ensure proper legal framework to support wCBDC's experimentations (paragraph 37).	High	Medium term
Gather feedback and empirical data to assess and quantify the impact of government bond tokenization and wCBDC on the capital market (paragraph 38).	Medium	Medium term

Table 1. Key Recommendations		
Recommendations for BSP	Priority	Timeframe 1/
Product-Market Fit and Suitability		
Define clear policy goals and develop initial hypotheses for wCBDC that could be tested and evaluated for its economic impact (paragraph 39).	High	Near term
Engage stakeholders in the co-design of a wCBDC to ensure it delivers meaningful benefits to users and supports practical, real-world applications (paragraph 40).	High	Medium term
Explore appropriate incentives and pricing structures to promote sustained adoption (paragraph 41).	High	Medium term
Research technical and economic opportunities that tokenization could concretely bring to the Philippine financial markets (paragraph 42).	Medium	Medium term
Technology		
Collaboratively undertake research on technology solutions for wCBDC and tokenization experimentations (paragraph 43).	Medium	Near term
Continue researching and implementing measures to ensure appropriate management of related digital risks, such as cyber security and operational resilience (paragraph 44).	High	Medium term
Continue to improve upon technical shortcomings, deficiencies, and risks identified in the Project Agila's sandbox (paragraph 45).	High	Medium term

1/ Near term: < 12 months; Medium term: 12 to 24 months.

I. Executive Summary

The IMF conducted a hybrid technical assistance mission on wholesale CBDC (wCBDC) to the Bangko Sentral ng Pilipinas (BSP) between November 26 and December 4, 2024. The mission intended to support the BSP in its study and experimentation to determine whether wCBDC could effectively address challenges within the Philippines' wholesale payment landscape. The mission consisted of two components: the facilitation of stakeholder engagement through an IMF-led design-thinking workshop, and the development of a preliminary roadmap for the future exploration of wCBDC.

The onsite mission's stakeholder workshop, facilitated by the mission team, engaged key representatives from financial institutions, government and regulatory agencies, and financial market infrastructure. Through these interactions, the BSP sought to understand the current landscape of wholesale payments, identify pain points, and explore innovative solutions that wCBDC could offer. The workshop participants noted challenges such as inefficiencies in settlement processes and a lack of interoperability among existing payment systems. The mission team and BSP then employed the insights gained from stakeholders to prioritize potential use cases for the next phases of Project Agila's wCBDC experimentation. Two use cases stood out for future consideration: settling of tokenized government bonds and enhancing cross-border payments.

A key outcome was the development of a preliminary roadmap to guide the BSP's exploration of wCBDC based on the identified use cases. This roadmap emphasized a multi-phase approach, structured around the IMF's "5P methodology". Concretely, the roadmap presented four action items under the foundational workstream, including resource allocation, identification and alignment of policy goals, continuous stakeholder engagement, and suitability assessment. The foundational action items were then followed by research and experimentation activities for the two identified use cases of tokenized government bonds and cross-border payments.

The mission presented several recommendations to ensure a successful implementation of the roadmap. First, the mission recommended the BSP to allocate resources effectively and engage multi-disciplinary expertise within the dedicated working teams, while also identifying overlapping action items and dependencies for each use case. Also, the mission emphasizes the importance of establishing KPIs and prioritizing digital risk management, particularly in the realm of cybersecurity. Moreover, it recommended the development of a robust legal framework to support wCBDC technical experimentation and potential issuance, alongside exploring incentives for sustainable adoption of wCBDC. Also, ongoing engagement and collaboration with stakeholders are crucial to ensure that the design of wCBDC would deliver benefits to the users and support real-world applications. Lastly, the mission encouraged the BSP to refine the projects and the roadmap continuously, drawing on feedback from the stakeholders and adapting to evolving market conditions. Collectively, these recommendations and strategies aim to ensure that, should the BSP decide to issue wCBDC, it will effectively address the needs and priorities of the financial system and contribute positively to the Philippine economy.

II. Introduction and Background

1. **The BSP has been exploring CBDC since 2020 and pursuing a wCBDC prototype development since 2022 under “Project Agila”.** Developed by Oracle Philippines within a BSP-led sandbox, the project involves selection of financial institutions (FIs) to enhance capacity and knowledge of wCBDC for the BSP and the financial industry. Insights from the project will guide how wCBDC technology could address payment challenges in the Philippines.
2. **The first sandbox, which was the initial Project Agila prototype, tested wCBDC for interbank transfers and 24/7 operations.** The wCBDC testing operated on a distributed ledger technology (DLT), which also allowed for the issuance of tokenized securities. The testing focused on assessing the performance and functionalities of the technology in supporting various stages of the wCBDC lifecycle and conducting interbank transfers. The IMF mission team conducted a desk review of the outcome reports in late 2024 (see details below and in Annex IV).
3. **In the preceding three CBDC missions to the BSP, the mission team provided capacity development support that laid the groundwork for this mission.** In the first mission (June 2022), it delivered a virtual and introductory training on CBDC, followed by initial assessments of potential wCBDC use cases (December 2022). The third phase provided policy, legal, and project management guidance in preparation for the wCBDC prototyping phase (November 2023).
4. **The third mission (November 2023) recommended that the BSP conduct a stakeholder engagement session to ensure wCBDC goals, use cases, and designs are aligned with the broader financial system stakeholders.** This recommendation was the basis of this fourth mission.
5. **The BSP sought to explore additional use cases for wCBDC with private and public-sector stakeholders and to construct a roadmap for future wCBDC exploration.** A consultation with the industry was expected to help inform which use cases of wCBDC the BSP could prioritize given the challenges in the Philippines’ financial markets and payment systems. The roadmap would target a medium to long-term horizon and inform the BSP’s actions for the wCBDC exploration.
6. **The mission aimed to facilitate the BSP in collecting inputs for its wCBDC roadmap, through four major activities:**
 - First, the mission team delivered the presentations on key wCBDC issues, from policy to technology, notably from the IMF’s recent CBDC Virtual Handbook and tokenization projects.¹
 - Second, the mission team conducted a desk review of the BSP’s recent sandbox testing.
 - Third, the mission team supported the BSP in conducting a three-day workshop with stakeholders to prioritize wCBDC use cases using a human-centric design approach.
 - Fourth, the mission supported the BSP’s development of a preliminary roadmap for wCBDC research and exploration.
7. **This report is organized as follows.** Section II serves as background information. Section III summarizes key findings from the stakeholder workshops. Section IV outlines key components from the preliminary wCBDC roadmap and recommendations. Section V concludes. Annexes include the detailed roadmap, the overview of the Philippines’ wholesale payment landscape, the workshop methodology, ongoing wCBDC projects and use cases around the world, the summary of the desk review of Project Agila’s reports, and the medium-term capacity development workplan.

¹ See [the CBDC Virtual Handbook](#) for a complete list and details of its chapters.

III. Findings from Stakeholder Engagement

8. This section presents an overview of the inputs and key findings from the stakeholder engagement workshops during the mission. The workshops were held for two days with participants including nine FIs from Project Agila, representatives from the Bankers Association of the Philippines, and officials from various government agencies such as the Securities and Exchange Commission (SEC) and the Bureau of the Treasury (BTr). Key players in financial market infrastructure (FMI) were also involved, including the Securities Clearing Corporation of the Philippines (SCCP), the Philippine Depository & Trust Corporation (PDTTC), Philippine Payments Management, Inc. (PPMI), and the Philippine Dealing & Exchange Corporation (PDEX). Additionally, the BSP staff attended as technical observers and provided an overview of the Philippines' wholesale payment system (Annex II).

A. Objectives and Approach

9. The primary objective of the stakeholder engagement was to inform the next potential use cases for wCBDC and Project Agila's roadmap. The participants shared their experiences and perspectives about priorities, needs, and challenges in the wholesale payment systems. This enabled the BSP to better understand the evolving financial landscape and how wCBDC may play its role. It is important to clarify that the workshop does not serve as a comprehensive assessment of financial market infrastructures (FMIs) in the Philippines. Rather, it provides a platform for market participants to directly highlight their views about areas of improvement for BSP's awareness and consideration.

10. The mission adopted a human-centered design approach when facilitating the workshops. This approach was selected for its ability to structure and facilitate meaningful dialogue among a diverse range of participants. It also highlighted the BSP's interest in fostering cooperation with stakeholders about wCBDC-related innovation. The approach helped to identify root causes, swiftly gathering feedback on preferences across competing exploration areas. It also fostered collaborative consensus-building through multi-disciplinary perspectives. This process aimed to inform the BSP's subsequent steps in its wCBDC roadmap (Annex IV).

Methodology

11. The process consisted of two key activities: 1) a two-day stakeholder engagement workshop to discuss key challenges in wholesale payments and to explore wCBDC as a solution, and 2) a collaborative session between the BSP and the mission team to draft the preliminary wCBDC roadmap. The high-level summary of the two workshops is illustrated in Figure 1 and the detailed process and inputs from the workshop are outlined in Annex III:

- *Stakeholder engagement:* The mission team analyzed Project Agila's results, focusing on pilot participants' pain points. On the first day, participants validated and reflected on key pain points and assessed existing and planned solutions. Then, they identified gaps and potential wCBDC use cases. On the second day, they explored value propositions, risks, additional relevant stakeholders, and feasibility of these use cases based on their impact and feasibility.
- *Preliminary roadmap:* By adopting a "Future Thinking" method, the mission helped the BSP build a future vision for the identified use cases. The method enabled the BSP to establish a long-term direction and vision of wCBDC for the Philippines. Both teams collectively designed the roadmap, which consisted of action items covering various aspects of the wCBDC use cases, strategy, technology, operations, and macro-financial implications. Finally, the BSP engaged in discussions to prioritize the identified action items for both the short and medium terms.

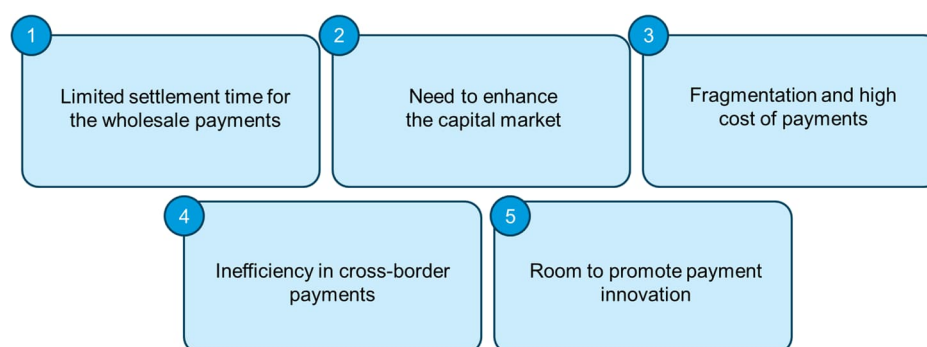
Figure 1. High-Level Process of Stakeholder Engagement and Roadmap Workshops

Day 1 with stakeholders		IMF	Day 2 with stakeholders		IMF & BSP Roadmap workshop
Why is wCBDC needed?	What are some solutions mitigating the challenges?	What are the gaps/persistent challenges?	What are potential use cases and value propositions for each?	What use cases could be prioritized?	What is the vision of the use case and the plan to achieve it?
Validate current pain points and gain a deeper understanding on the practical issues and implications for stakeholders regarding those challenges	Map existing and planned market-level initiatives that are contributing to addressing the identified pain points. Assess the perceived effectiveness of these initiatives in mitigating the challenges faced by the stakeholders.	Analyze the results of stakeholder discussions to identify persistent gaps between existing pain points and the solutions already deployed or planned. Assess the areas where these gaps could be effectively addressed by wCBDC and outline an initial set of potential use cases for further exploration with authorities and stakeholders.	Reflect on the risks and benefits of each use case and translate these insights into actionable steps. Identify specific questions to be investigated and stakeholders to involve in further exploration.	Prioritize the most promising use cases for further exploration based on the identified pain points, existing initiatives, and the potential impact on the country.	Visualizing the future of wCBDC to define specific milestones and identify actionable steps to realize our vision for a transformative wCBDC.

B. Key Challenges in the Wholesale Payment Landscape Identified by the Participants

12. The mission summarized key high-level challenges in the Philippines' wholesale payment landscape, captured from the BSP's initial survey with Project Agila's participants during the sandbox testing. These challenges were discussed with the BSP prior to the mission's workshops and served as a foundational reference for the stakeholder engagement. The challenges, listed in Figure 2, were categorized as follows: 1) limited settlement time for the wholesale payments, 2) need to enhance capital markets development, 3) fragmentation and high costs of payments, 4) inefficiency in cross-border payments, and 5) room to promote payment innovation.

Figure 2. High-Level Challenges Captured from Project Agila's Survey



13. The identified challenges from Project's Agila's survey serve as a reference point for the workshop participants to elaborate and validate the effectiveness and gaps of current initiatives. The identified challenges act as a foundational framework during the workshop, guiding participants to have in-depth discussions on their respective fields or sectors. The inputs were then used as a basis for developing potential use cases for wCBDC.

14. Below is a summary of the premises of the aforementioned challenges, the feedback from workshop participants, and the current or planned initiatives to tackle them.

▪ *Limited settlement time for the wholesale payments:*

- *Premise:* PhilPaSS^{plus} is the Philippines' real-time gross settlement system (RTGS) for processing and settling large value funds transfer between FIs, including settling government and corporate bond transactions. It also supports settlement of transactions from InstaPay, PESONet, checks and ATM for individuals and businesses. InstaPay and PESONet are automated clearing houses (ACHs) created for the Philippines' retail payment system. InstaPay is an instant payment system, launched in April 2018, which facilitates real-time, 24/7, low-value payments using an account number, QR code, phone number, or email address via banks accounts and e-wallet providers. PESONET also facilitates payments from bank accounts or e-money wallets, but transactions are processed in batches and operates only during business hours. Both ACHs rely on their respective Clearing Switch Operators (CSOs) to submit the net settlement information to PhilPaSS^{plus} for direct settlement.²

While payees receive the money via InstaPay immediately, its settlement is on a deferred net basis and the respective bank assumes the settlement risk until the transaction is fully settled in PhilPaSS^{plus}.³ Particularly during outside of regular operating hours, banks generally maintain liquidity buffers in their Demand Deposit Accounts (DDAs) in PhilPaSS^{plus} to ensure they have adequate balances for settlement until the system opens.⁴

- *Feedback from the workshop:* Many participants did not show strong interest in 24/7 PhilPaSS^{plus} operation, voicing concerns of availability of resources to ensure continuous payment and settlement operations and compliance. Nevertheless, some expressed interests in a payment system that could allow for lending and borrowing during off hours, such as emergency lending in case of shortfalls of funds, particularly for FIs with international operations. In terms of resiliency, PhilPaSS^{plus} can continue operation with its backup site and business continuity plan (BCP) during unexpected disruptions (e.g., typhoons). Nevertheless, some participants were concerned about an over-reliance on a single technology and an external vendor, raising a risk of single-point-of failure.
- *Assessments of key existing/ planned solutions:* The BSP is discussing with the market to assess the need for extending the operating hours of PhilPaSS^{plus}. While the system has been effective in withstanding disruptions, some market participants shared that the BSP should continuously seek industry feedback on the BCP and operating guidelines to better align with market needs and conditions.

▪ *Need to enhance the capital market:*

- *Premise:* Despite significant growth in the Philippines' capital market, the sizes of the Philippines's government bond and equity markets per nominal GDP remain small relative to some regional peers, pointing to potential for growth (see Figure 3 and 4).

² Source: [BSP's National Retail Payment System](#).

³ Source: [Philippines' Financial Sector Assessment Program Technical Note: Retail Payments](#).

⁴ PhilPaSS^{plus} currently operates from 9:00 am to 5:45 pm from Mondays through Fridays, except for national holidays.

- **Bonds:** The domestic bond market in the Philippines has been dominated by government bonds. The total outstanding government bonds (including treasury bills and bonds), growing at 9.11 percent year-on-year, reached PHP 10,930 billion at the end of 2024, accounting for 40.8 percent of the total local currency bond market. The government bonds are issued through an auction process by FIs that are designated as primary dealers called government securities eligible dealers (GSED). The secondary market for the government bonds generally lacks depth and liquidity, and the interbank repo market is inactive on most days.⁵ In addition, corporate and central bank bonds represented 5.13 and 2.96 percent of total local currency bond market, respectively.⁶

Retail treasury bonds accounted for 28.7 percent of the local currency market at the end of 2024. They typically have long maturity and are seen as stable funding sources. In terms of investors, the Philippines' retail bond program targets a broad spectrum, including not only individual investors, but also non-bank investors such as insurance companies, mutual funds, and large corporations. Even though these large investors could access regular government bonds through GSED, they often prefer to access the retail bond issuance channel for its simpler acquisition process and larger issuance sizes.⁷ Banks and investment houses consistently represent the largest investor group.

⁸As a result, individual investors hold only a small share.⁹

Bureau of Treasury (BTr) oversees the issuance and servicing of government bonds. National Registry of Scripless Securities (nRoss) and the Philippines Depository Trust Corporation (PDTC) handle registrations, clearing, and settlement of government bonds and other fixed income securities, respectively. Both agencies have direct access to PhilPaSS^{plus} to settle bond payments in central bank money. The Philippine Dealing Systems Holdings Corporation (PEDX) facilitates the secondary bond market trading.

- **Equities:** Retail investors constitute 98.5 percent of stock market accounts in the Philippines in 2023, with 17.6 percent classified as active accounts.¹⁰ Since 2000, the country has seen the lowest number of newly listed firms and capital raised through initial public offerings (IPOs) compared with its ASEAN peers. The lengthy listing process, combined with stringent requirements and the complex fee structure, has contributed to this stagnation. Furthermore, the burden of IPO-related taxes has diminished the attractiveness of raising funds through the equity market for companies.¹¹

The Securities Clearing Corporation of the Philippines (SCCP) serves as a Central Counterparty (CCP) for trades executed at the Philippines Stock Exchange (PSE). Similarly, the equities are settled in a delivery-versus-payment (DvP) model 3 with settlement cycle of T+2 on a multi-lateral net-settlement basis, improving from T+3

⁵ Source: The IMF/WB TA report—Strengthening the Local Currency Government Bond Market in the Philippines (July 2024).

⁶ Source: [Asian Bonds Online](#).

⁷ Source: The IMF/WB TA report: Strengthening the Local Currency Government Bond Market in the Philippines (July 2024). In addition, the report cited that only 0.025 percent of the adults with bank accounts invested in retail government bonds.

⁸ Source: [Asian Development Bank 's Asia Bond Monitor November 2024](#).

⁹ Source: [Bureau of the Treasury \(BTr\)](#).

¹⁰ Source: [the Philippines Stock Exchange Report - Stock Market Investor Profile 2023](#).

¹¹ Source: [OECD Capital Market Review of the Philippines 2024](#).

since August 2023.¹² The DvP 3 model generally entails a settlement risk due to the time delay between the payment and the security delivery.¹³ Unlike government bonds, the cash leg is settled in commercial bank money via ten broker settlement banks. The SCCP and system participants are thus exposed to both credit and liquidity risks due to their reliance on settlement banks.

Figure 3. Local Currency Bond Market as Percent of Nominal GDP ^{1/}

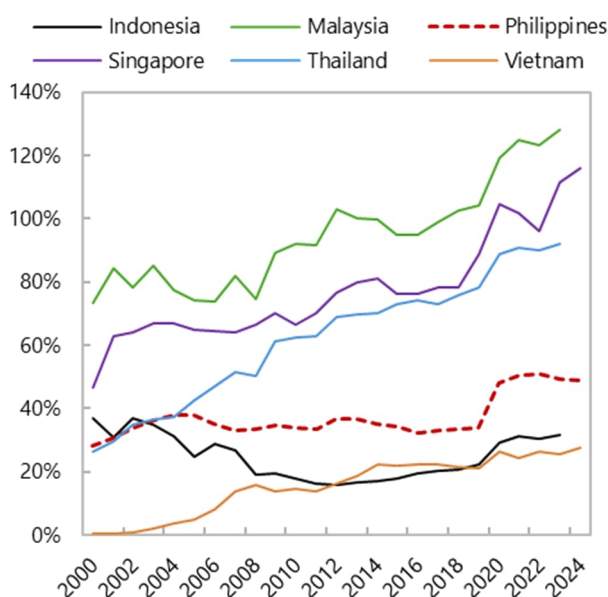
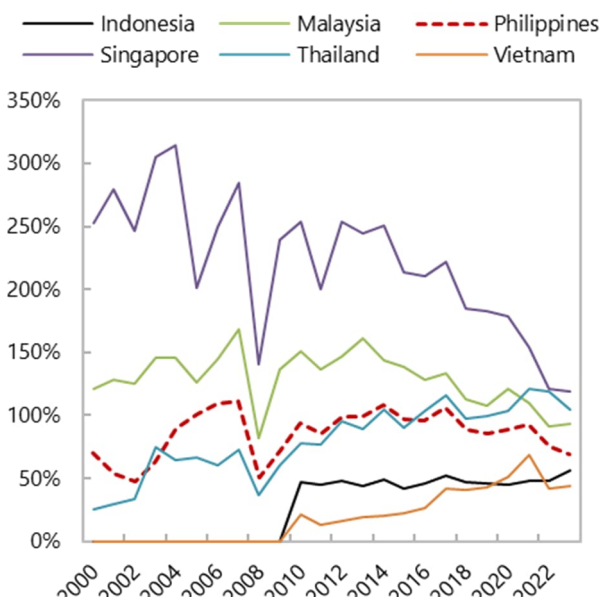


Figure 4. Stock Market Capitalization as Percent of Nominal GDP



Sources: Asian Bond Online, CEIC, IMF's General Government Gross Debt, and IMF staff calculations.

1/ The bond market includes government, corporate, and central bank bonds

- *Feedback from the workshop:* Both participants and the BSP reflected the need to improve efficiency and enhance accessibility in securities transactions. They acknowledged that raising funds in the capital market generally incurs high costs due to the involvement of numerous intermediaries and complex compliance layers. Furthermore, they highlighted the importance of deepening capital markets by diversifying the investor base and enhancing secondary market trading. For the local bond market, encouraging more participation from individual investors could also provide BTr with a more stable source of financing, while promoting financial inclusion and enhancing accessibility of savings instruments.

Additionally, SCCP prefers to maintain its relationship with broker settlement banks, finding it more cost-effective than accessing PhilPaSS^{Plus} directly. Currently, the settlement banks charge no fees to SCCP for equity settlements, unlike the fees that

¹² Source: [SCCP](#).

¹³ DvP Model 1 settles securities and funds on a gross basis, with simultaneous delivery and payment. DvP Model 2 settles securities grossly throughout the cycle but funds net at the end. DvP Model 3 settles both legs on a net basis at the end of the cycle. Each model carries distinct risks and implications. Source: [CPSS](#).

SCCP would incur if it settles directly through PhilPaSS^{Plus} access. Additionally, the banks noted their recent investment in clearing and multi-lateral netting software from the London Stock Exchange Group, emphasizing that features like liquidity savings may not be easily replicable in the PhilPaSS^{Plus} environment.

○ *Key existing/ planned solutions:*

- *Adoption of new technology and digital solutions:* BTr has implemented several digitalization initiatives to allow more individuals to invest in retail treasury bonds and treasury bills at a primary market offering. This can be done via BTr's website, mobile banking apps, BTr's mobile app "Bond.ph", or Philippine Digital Asset Exchange (PDAX) which normally operates as a platform for cryptocurrency and digital assets trading. The minimum purchasing amount has also been reduced from PHP5,000 to PHP500 (about 88 USD to 8.8 USD). There are also plans to allow electronic money issuers (EMIs), such as GCash, to offer such service to their clients. In addition, BTr launched an early form of tokenized treasury bonds (TTB) in 2023 which utilized permissionless, public blockchain for a bond purchase registry for targeted institutional investors (Box 1).
- *Capital Market Development Council (CMDC):* CDMC serves as a key institution aimed at advancing the country's capital market. It comprises a diverse group of participants, including government agencies like the SEC, BTr, and BSP, alongside representatives from private sector entities such as banks, investment houses, and other FIs. The CMDC focuses on developing policies and initiatives to enhance market liquidity, investor confidence, and overall market efficiency, thereby contributing to the sustainable economic growth of the Philippines.

Box 1. BTr's Tokenized Treasury Bonds

In November 2023, BTr, in partnership with PDAX, LandBank of the Philippines, and the Development Bank of the Philippines, launched an initial foray into Tokenized Treasury Bonds (TTBs) for qualified institutional investors. TTBs offered an annual fixed rate of 6.5 percent, with one-year maturity. Due to a strong demand from the investors upon its issuance, BTr eventually increased the original IPO amount of TTBs from PHP10 billion to PHP15 billion. While the Treasury bonds are not represented and traded on a DLT as the term "tokenization" may commonly refer, their purchase information was recorded on a DLT in parallel to the existing process.

Source: BTr and PDAX

- *Fragmentation and high costs of payments:*
 - *Premise:* Fragmentation in payment systems typically refers to the presence of multiple, incompatible payment networks, standards, or providers, which can obstruct seamless transactions. Intermediaries that incur costs while navigating these disparate systems may ultimately pass these expenses on to users, resulting in higher payment fees for end consumers. For example, retail payments processed between PESONet and InstaPay incur fees due to the involvement of two different CSOs for cross settlement – Bancnet, Inc. and the Philippine Clearing House Corporation (PCHC), respectively. For end users the retail transaction fees in the Philippines are higher on average compared with its regional peers. While other regional countries typically do not impose fees for small,

person-to-person transactions, most commercial banks in the Philippines typically charge around PHP10-25 per transaction, while the fees become higher for corporate clients – reaching up to PHP500 per transaction (Table 2 **Table 2. Retail Payment System Fees in the Region**). Nevertheless, digital banks and some EMIs do not charge any fees, opting to absorb these costs to encourage adoption.¹⁴

Table 2. Retail Payment System Fees in the Region

Country	Retail Payment System	Transfer Out Fees
Indonesia	BIFAST	Free to USD0.15
Malaysia	DUITNOW	Free to USD0.11
Philippines	INSTAPAY	USD0.14 to USD0.50
Singapore	PAYNOW	Free
Thailand	PROMPTPAY	Free to USD0.17
Vietnam	NAPAS	Free to USD0.22

Source: [Payments and Commerce Market Intelligence](#).

- *Feedback from the workshop:* Some participants raised concerns regarding the IT integration challenges posed by different payment systems with varying technology standards, thus creating obstacles to achieving seamless and cost-efficient transactions. This complexity and the associated costs have also led to an increased likelihood of failed transactions. As a result, many commercial banks have chosen not to directly connect to every payment system, instead prioritizing specific systems.
- *Key existing/ planned initiatives:*
 - *Consolidations of CSOs:* Given the current setup with two CSOs and two ACHs, some participants suggested that a consolidation of the two CSOs could allow for more standardization of IT, operations, and compliance, thus leading to lowering fees.
 - *Integration of ISO 20022:* The adoption of ISO 20022 standardized messaging protocol will facilitate more efficient and effective communication between FIs. This transition is expected to improve interoperability and reduce payment transaction costs. The BSP has been actively promoting this integration, encouraging FIs to adopt the standard to support the modernization of payment systems in the country. Many FIs are in the process of integrated ISO 20022 and are mostly near completion.
 - *Sharing IT resources:* For example, the BSP has provided IT resources to nRoss to ensure smooth integration and operations with PhilPaSS^{plus}.
 - *Bills Pay PH:* Bills Pay PH is an interoperability facility to streamline bill payments, allowing individuals and businesses to pay their bills such as utility, rent subscription, and credit cards with no service fees. With this facility, a customer can make payments to a biller who has no existing arrangement with the customer's payment service provider.¹⁵
- *Inefficiency in cross-border payments:*

¹⁴ Source: [BSP's Summary of Corresponding Transfer Fees through Digital Channels](#).

¹⁵ Source: [Bills Pay PH - FAQs](#).

- *Premise:* Wholesale cross-border payments in the Philippines remain rife with common issues, such as limited access, high costs, low speed, and lack of transparency. These issues stem from the involvement of multiple intermediaries, varying communication and data standards, cross-country regulatory compliance, as well as operating time zones. Currently, a small number of correspondent banks dominate the space. As for retail cross-border payments, the Philippines is one of the largest global recipients of remittances. Anecdotal evidence gathered in a previous TA mission suggests that some individuals have been using crypto-assets to send and receive money from abroad, although some individuals use crypto-assets for speculative purposes.¹⁶ According to a Chainalysis report in 2024, the Philippines ranks eighth in the Global Crypto Adoption Index and seventh among countries in Central and Southern Asia and Oceania for the value of crypto assets received.¹⁷
- *Feedback from the workshop:* While most participants agreed on the existing challenges in cross-border payments, some added that smaller banks faced challenges in competing with nonbanks or fintech companies who may better leverage crypto assets to offer cross-border payment services with less regulatory restrictions. In addition, some participants expressed interest in reducing reliance on SWIFT to reduce associated costs.
- *Key existing/ planned solutions:*
 - *BSP's participation in cross-border initiatives:* The BSP is an observing member of Project mBridge, a cross-border wCBDC experimentation project that has shown the feasibility of rapid cross-border payment transfers.¹⁸ However, many central banks, including the BSP, are adopting an observer status to potentially join as full participants while minimizing resource commitments and geopolitical challenges. For retail cross-border payments, the BSP, along with the central banks of India, Malaysia, Singapore, and Thailand, is participating in Project Nexus, which aims to develop a blueprint for connecting instant payment systems across countries.¹⁹ In addition, the BSP is part of Regional Payment Connectivity initiative (RPC), which uses a single QR code to link and enables retail cross-border payments for other ASEAN countries.
 - *Harmonized ISO 20022 data requirements for cross-border payments:* As outlined in the recent BIS report, market participants are encouraged to begin preparations and to foster the implementation of the harmonized ISO 20022 data requirements for cross-border payments messages by 2027.²⁰
 - *Room to foster payment innovation:*

¹⁶ Source: The IMF TA report–Philippines's Wholesale Central Bank Digital Currency Use Case Analysis (July 2023).

¹⁷ Source: [Chainalysis' 2024 Geography of Crypto Report](#). The period studied for crypto asset value received was from July 2023 – June 2024.

¹⁸ Project mBridge experiments with a multi-wCBDC common platform for cross-border payments. It seeks to solve some of the key inefficiencies of cross-border payments, such as high costs, low speed and transparency, and operational complexities. Currently, there are five full participants — the Bank of Thailand, the Central Bank of the United Arab Emirates, the Digital Currency Institute of the People's Bank of China, the Hong Kong Monetary Authority, and Saudi Central Bank. Source: [BIS](#).

¹⁹ Project Nexus is an initiative designed to enhance cross-border payment efficiency by linking national instant payment systems and creating a standardized set of protocols and governance frameworks. The pilot phase of Project Nexus involves Singapore, Malaysia, and Thailand, with plans to expand to other countries. These nations are leveraging their advanced instant payment systems, such as PayNow and PromptPay, to demonstrate the feasibility of interconnected cross-border payments. Source: [BIS](#).

²⁰ Source: BIS report ["Harmonised ISO 20022 data requirements for enhancing cross-border payments"](#).

- *Premise:* Participants of Project Agila demonstrated a strong enthusiasm for innovation, driven by advancements in technologies, rising demand for digital financial services, competitive pressures, and regulatory support from the BSP. The participants highlighted opportunities related to wCBDC and tokenization, including real-time foreign exchange and securities settlement, as well as real-time conditional payments.
- *Feedback from the workshop:* The rise of crypto assets has created competitive pressure on FIs to adapt while ensuring compliance. Despite interests in offering innovative financial products and services, FIs have so far taken a cautious approach. This is in part due to certain regulatory uncertainty with respect to products and services related to crypto-assets, blockchain applications, and tokenization of securities. They also noted that clarity regarding the BSP's policy direction and the development of wCBDC would potentially influence how the market for tokenization would take shape.
- *Key existing/ planned solutions:*
 - *Gaining access for nonbanks to PhilPaSS^{plus}:* PhilPaSS^{plus} has already allowed direct access for nonbanks with quasi-banking functions, nonbank EMIs, and government agencies that participate. It has aims to promote an inclusive national payment system and to allow nonbanks to make efficient and low-risk fund transfers without relying on sponsoring PhilPaSS^{plus} participants. As of now, there has been only one nonbank EMI, OmniPay, that has gained direct access to PhilPaSS^{plus}.
 - *Virtual Asset Service Providers (VASP):* The BSP issued licenses for VASPs in 2020 to enhance consumer protection and mitigate risks associated with virtual assets, such as crypto-assets.²¹ As of January 2025, there are ten active VASPs (eight are nonbanks and two are banks), and three inactive VASPs.²² The BSP closed the regular application window for new VASP for three years from September 2022.²³
 - *Regulatory sandboxes:* The BSP introduced a regulatory sandbox framework in 2022 to test new financial technologies and business models while ensuring consumer protection and regulatory compliance. By June 2024, it had received 13 applications, primarily focused on blockchain innovations, with one approval for a pilot of PHP-backed stablecoin so far.²⁴ Following this, the SEC launched its own Strategic Sandbox to test innovative financial solutions related to the capital market.²⁵
 - *Open finance initiative:* In 2021, the BSP launched Open Finance Pilot to test open finance principles with partnered FIs, focusing on secure data sharing and use cases like personal finance management and credit scoring.²⁶ For next steps, the BSP aims

²¹ Source: [BSP's Circular No. 1108 – Guidelines for Virtual Asset Service Providers \(VASP\)](#).

²² Source: [VASP.pdf](#)

²³ Source: [BSP's MOU No. M-2022-035 – Modified Approach in the Grant of VASP Licenses](#).

²⁴ Coins.ph, a crypto exchange platform, has received approval by the BSP to conduct a pilot testing for a PHP-backed stablecoin under a controlled environment. The BSP will evaluate the stablecoin's viability and impact before considering further approval for real-world applications. Coins.ph has been previously granted VASP and EMI licenses. Source: [Coins.ph](#).

²⁵ Source: [Accelerating Innovation: The Philippines' Journey with Regulatory Sandboxes](#).

²⁶ Source: [BSP's Circular No. 1122 – Open Finance Framework](#).

to establish a comprehensive open finance framework that supports innovation while ensuring consumer protection and data security.

C. Stakeholder Opinions of Potential wCBDC Use Cases

15. The participants identified potential use cases of wCBDC, which could help address the existing challenges. The participants noted that inefficiencies in cross-border payments present the most significant gaps for improvement, followed by a need to further develop the capital market. While the challenges need to be addressed by fostering more innovation, improving fragmented payment systems and limited operating hours of the wholesale payment systems appear to have relatively more effective solutions in either existing or planning stages (Annex IV, Annex IV. [Figure 2.](#)).

16. It is important to note that this does not suggest wCBDC as a more effective solution than alternatives, nor could it solely solve all the identified challenges. Rather, it highlights an avenue for the BSP and the industry to further explore wCBDC. The BSP should continue to pursue alternative solutions to address those challenges while exploring wCBDC use cases.

17. The identified wCBDC use cases include:

- *wCBDC for settling tokenized government bonds:* While significant progress has been made in enhancing the capital markets, challenges remain in enhancing secondary market trading and improving access for individual investors and other non-primary dealer FIs. While testing the use case of tokenized securities could be applicable with a wide range of asset classes, the stakeholders believed it was important for wCBDC to support BTr's ongoing experimentation with TTBs. wCBDC would serve as a tokenized cash leg for settling these tokenized government bonds. By tokenizing both bonds and cash, the use case enables the BSP to assess the benefits of wCBDC as cash settlement for tokenized assets. Some promising features of tokenization and wCBDC that warrant exploration include:
 - *Simultaneous and conditional DvP settlements in central bank money:* As a cash leg for settling tokenized bonds, wCBDC transactions could significantly lower principal, liquidity, credit, and settlement risks compared to commercial bank money. Both the bonds and cash legs can be programmed for simultaneous and conditional DvP settlements, further mitigating these risks.²⁷ Furthermore, integrating smart contracts can automate and streamline the bond's transactional processes, such as coupon and dividend distribution, auctions, mark-to-market valuations, and ownership transfers.
 - *Flexible fractionalization of tokenized securities:* Tokenization can facilitate flexible fractionalization, enabling any amount of an asset to be traded based on investors' needs, without being constrained by minimum investment requirements. Additionally, it could allow for more dealers to be involved in distributing tokenized bonds to individual investors. This capability can lower investment barriers and broaden access for a diverse range of investors, aligning with BTr's objectives. Also, it may enhance bond liquidity and stimulate secondary market trading, as small investors can easily buy and sell bonds at greater ease. While there have already been efforts to reduce minimum investment

²⁷ The process of simultaneous settlement of securities is often referred to as atomic DvP, which guarantees that the delivery of securities and the corresponding payment occur at the exact same time, and neither party can complete their part of the transaction without the other party simultaneously fulfilling their obligation. [Schär \(2024\)](#) argues that true atomicity can only be achieved under a single base layer.

amounts in bonds, the BSP should assess whether flexible fractionalization could further improve accessibility and if wCBDC is necessary to effectively support this feature.

As for equities, tokenization could potentially address certain frictions in the equity lifecycle. By reducing the reliance on intermediaries and streamlining equity servicing process such as dividend distribution and trading, transaction costs could be significantly lowered. Moreover, simultaneous equity settlement with wCBDC could support the movement from the current DvP3 model to the DvP1 model as well as address counterparty risks.²⁸ To achieve this, SCCP would need to establish a direct wCBDC account with the BSP. However, SCCP prefers to keep its existing commercial bank relationships despite being offered direct access to PhilPaSS^{plus}. This situation is unlikely to change even with the introduction of wCBDC. The BSP could further explore this matter and first encourage SCCP to utilize PhilPaSS^{plus} for settling its cash leg in central bank money, to be in line with Principles of Financial Market Infrastructures (PFMI), before considering the use case of wCBDC for equity settlement.²⁹

- *wCBDC for cross-border payments:* The participants expressed strong interest in further enhancing efficiency of wholesale cross-border payments, as effective solutions to this challenge have yet to emerge. This use case would require close collaboration with partner countries that engage in significant trade and investment with the Philippines. While most participants acknowledged the considerable potential benefits of these initiatives, many also recognized the complexities involved in the implementation as well as regulatory compliance. The capabilities of wCBDC that could potentially enhance cross-border payment efficiency, which the BSP could assess when testing this use case, include:
 - *Simultaneous and conditional payment-versus-payment (PvP) settlements in central bank money:* Under a platform shared between participating countries, cross-border wCBDC can facilitate near-instantaneous, cross-currency PvP settlements, significantly reducing the time and transaction costs as demonstrated by cross-border wCBDC initiatives such as Project mBridge and Project Dunbar. The solution could change the roles of corresponding banks, depending on the FX liquidity management models.³⁰ This speed can enhance cash flow for businesses engaged in international trade.
 - *Enhanced applications related to cross-border payments:* The increased speed and reduced costs associated with cross-border transactions using wCBDC could significantly enhance cash flow and improve liquidity for businesses involved in international trade and investment. Thus, there is an opportunity to explore extended applications of wCBDC, such as supporting payments for tokenized letters of credit or invoices. This innovation has the potential to enhance trade finance and supply chain management.
- *Additional RTGS functionalities with wCBDC:* To mitigate concerns about over-reliance on a single technology and enhance BCP for PhilPaSS^{plus}, wCBDC could be explored as a

²⁸ For more information on how tokenization could help with financial market inefficiencies, see [Agur, Villegas-Bauer, Mancini-Griffoli, Martinez Peria, and Tan, 2024](#).

²⁹ The issue was also raised by the mission team during the wCBDC TA in 2022. Principle 9 of PFMI indicates that an FMI should be settled in central bank money whenever it is applicable to minimize the credit and liquidity risk arising from the use of commercial bank money for settlement. Source: [BIS](#).

³⁰ In [Project mBridge's setup](#), the platform initiates netting of queued transactions periodically at the individual central bank level using an automated Liquidity Saving Mechanism. It is possible that in other platform model, a third-party operator like a corresponding bank could still play a role in managing liquidity and providing FX quotes.

complementary system. This could involve replicating additional RTGS functionalities with wCBDC, such as gridlock resolution and repo transactions.

18. In addition, the participants discussed key opportunities, challenges, relevant policy questions, and relevant stakeholders for future work. They also as evaluated and ranked potential impact and feasibility to implement for each of the identified use cases. It should be noted that these outcomes reflect only the opinions of the stakeholder workshop participants, who may have varying degrees of familiarity with the tokenization and wCBDC concepts (see Annex III for more details).

Box 2. Alternatives to wCBDC for Settling Tokenized Assets

One of the key motivations for central banks to pursue wCBDC is to provide a risk-free, central bank money for settling tokenized assets in the case where tokenization has reached significant adoption in the financial markets. While the private sector has been exploring innovative solutions that could also support the settlement of tokenized assets, such as tokenized bank deposits or stablecoins, these forms of payment would still be exposed to some credit risks from the issuing entities.

Most experimentations with wCBDC involve issuing a digital form of central bank money on a DLT-based platform for the payment and settlement of tokenized interbank transactions. However, two alternatives to wCBDC developed by the central bank stand out as providing safety to users without requiring the development of a new DLT-based platform or issuance of a new money by the central bank.¹ These are:

- *Trigger Solution:* Trigger solutions involve communication and a technical bridge that coordinate tokenized asset delivery in one platform (such as a privately developed DLT for tokenized asset trading) with payment and settlement in central bank money (reserves) on the RTGS. The two parts of the transaction are mutually conditional, where the payment is made only if the asset is transferred, and vice versa. The scheme can effectively enable settlement on a DvP basis of tokenized assets with reserves. Typically, a “synchronization operator” facilitates communication between the asset ledger and RTGS. The Bundesbank has been exploring a trigger solution by developing a technical bridge utilizing DLT to link market-operated DLT platforms to the Eurosystem’s traditional payment system. This allows for the settlement of DLT-based wholesale financial transactions in central bank money in the T2 service of TARGET Services.
- *Omnibus Account:* This scheme, initiated by the Bank of England in 2021 and operated by the Sterling Finality Payment System, facilitates payment made within privately operated tokenized asset platforms, with backing in central bank money. In this setup, an entity acting as a licensed FMI and operating a private DLT platform could hold the Omnibus Account maintained at the central bank. This entity then issues a digital representation of these funds on its DLT platform, which is then made available to participating FIs. When these FIs make tokenized asset transactions with one another, they use this tokenized money for payments. As a result, the on-chain tokenized money is available for payments, while the underlying funds are backed by and settled in central bank money via the Omnibus Account. The scheme also supports the use of a programmable DLT for DvP, Pvp, or other enhanced arrangements.

Sources: [Bank of England](#), [Bundesbank](#), [Cambridge Center of Alternative Finance \(CCAF\)](#), [Finality](#), and

¹ [In the IMF’s G-20 Note](#) which discusses different models of financial platforms and their macro-financial implications, trigger solution and omnibus account are aligned with ‘compatible ledger platform and ‘common ledger platform, respectively.

IV. wCBDC's Preliminary Roadmap

A. Objective and Approach

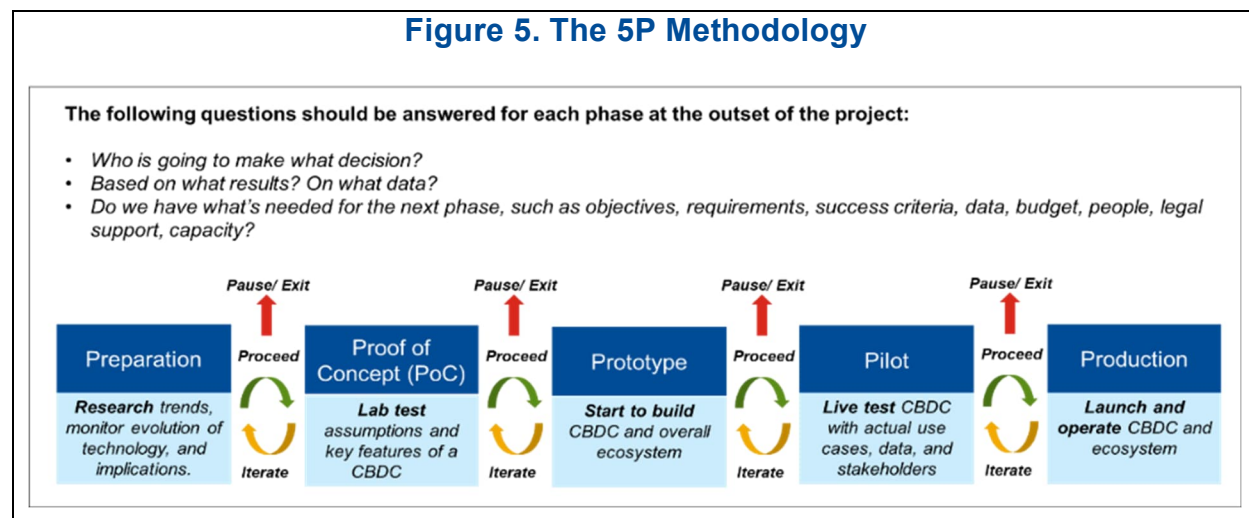
19. **The identified use cases provided the foundation for developing the preliminary wCBDC roadmap.** The mission conducted a workshop with the BSP to identify key action items, dependencies, milestones, and priorities. Similar to the stakeholder engagement workshops, the roadmap workshop employed a design thinking approach to ensure that the preliminary roadmap incorporates essential policy considerations as the BSP advances its wCBDC exploration.

20. **This roadmap represented a significant initial step in the BSP's wCBDC exploration, developed collaboratively by multiple BSP departments to ensure broad internal ownership and alignment across the organization.** Involving multiple departments from the outset helps foster shared goals and actions, strengthening commitment and facilitating smoother implementation of the roadmap. The inclusive process ensures that the BSP can effectively manage and adapt the initiative as it progresses.

5P Methodology

21. **The IMF introduced the “5P methodology” as a structured framework to support central banks in the research, experimentation, development, and operation of CBDCs.** This framework provides a phased, iterative process for managing experimental digital initiatives like CBDCs, ensuring that product development aligns with policy goals and practical use cases. The 5P methodology includes five key phases: Preparation, Proof-of-concept, Prototyping, Piloting, and Production. Each phase involves setting pre-defined criteria, goals, and governance preparations to determine whether to iterate within the same phase, to fully or partially advance to the next, or to pause or discontinue the project altogether (see Figure 5^{Figure 5}).³¹

Figure 5. The 5P Methodology



22. **Currently at the prototype phase, the BSP continues to explore potential use cases that could offer value to wholesale payment markets.** The stakeholder engagement was only one exercise

³¹ See [Tourpe, Lannquist, and Soderberg \(2023\)](#).

to identify potential use cases. The BSP would still need to conduct research to finalize use cases to be experimented, as well as their legal, technological, and macro-financial implications.

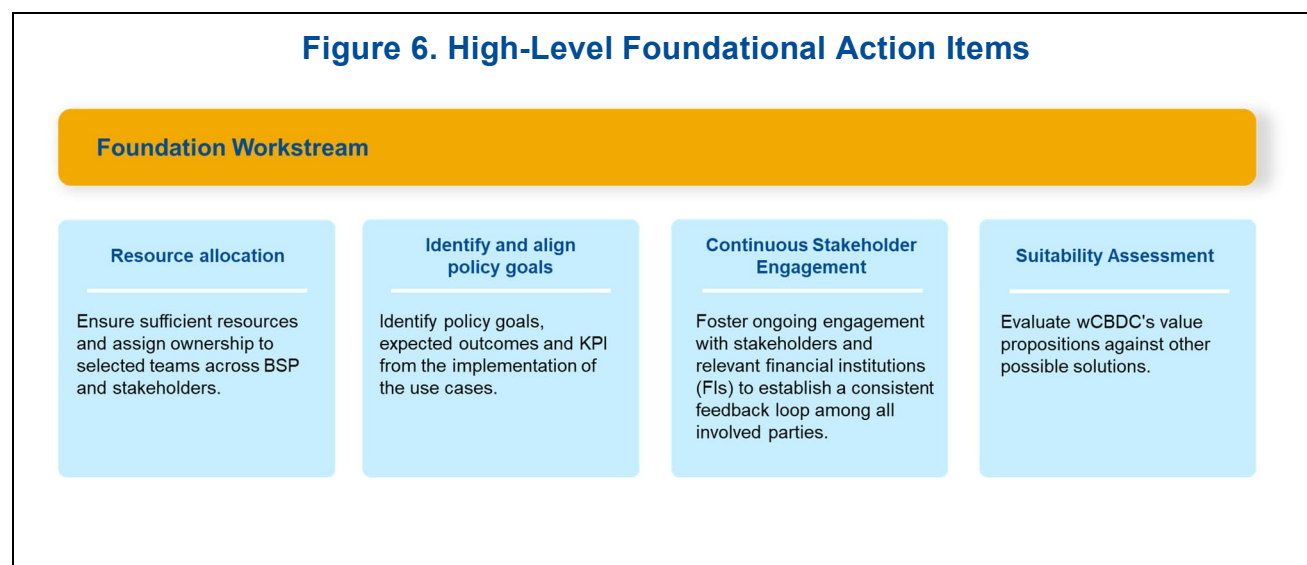
B. Preliminary Roadmap

23. As a result of the collaborative process, the preliminary roadmap was composed of three distinct workstreams. The first workstream focused on foundational actions necessary to support all wCBDC use cases, such as formalizing the team and securing resources. The second and third workstreams included the action items required for further experimentation and exploration of the two identified use cases—wCBDC for settling tokenized government bonds and cross-border payments. Each workstream included a prioritized list of action items aligned with the vision statement. Action items were grouped into short-term (1 year) and medium-term (2-3 years) timeframes. The detailed preliminary roadmap is presented in Annex I.

24. Due to limited participation and time constraints, the mission team and the BSP’s CBDC team decided to defer discussions of the use case of developing additional and duplicative RTGS functionalities for future review. This is because this use case, focused on replicating RTGS functionalities such as gridlock resolution or repo transactions with a wCBDC, would require less inter-organizational coordination and mainly rely on the BSP’s CBDC team to determine the next steps for sandbox testing. Both teams agreed to focus on two use cases that were deemed more complex and would necessitate alignment across multiple departments. The BSP CBDC team acknowledged the need to address all three use cases when refining the preliminary roadmap.

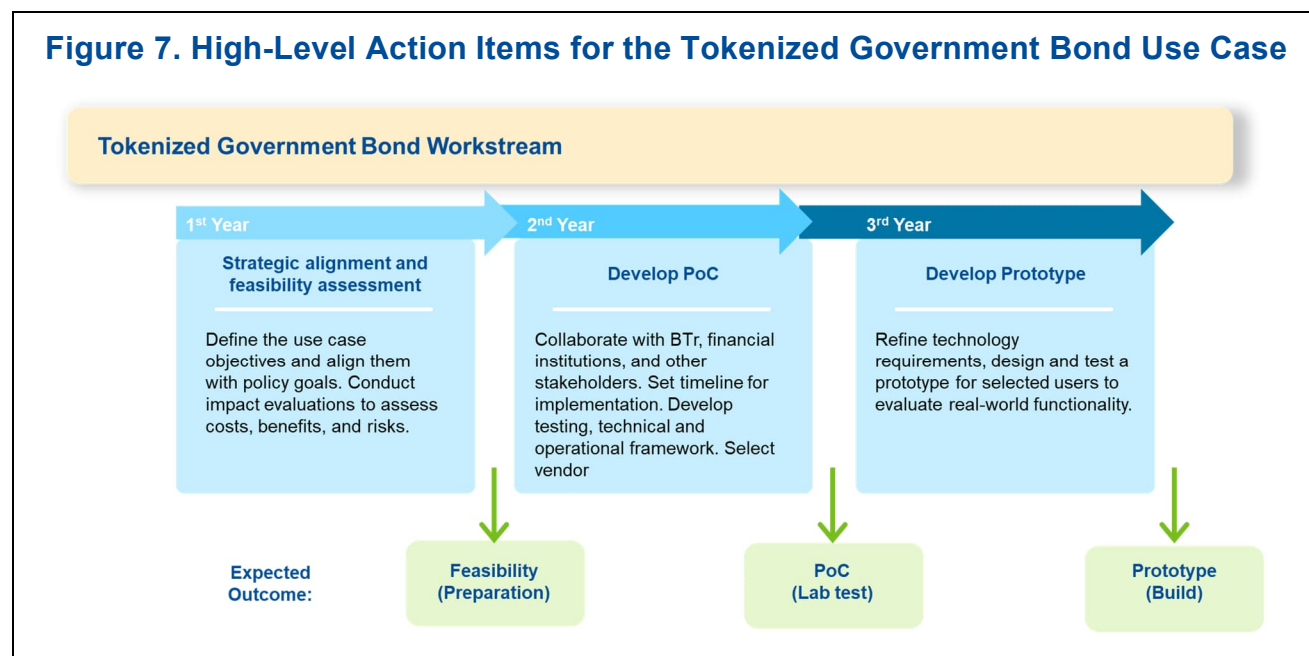
25. The foundation workstream should cover all essential activities that must be completed before initiating either of the two use cases. These should include resource allocation and team setup—both critical for determining the timeline of roadmap activities, as they directly depend on available resources. Additionally, each use case should clearly identify its policy goals and expected outcomes to enable progress tracking against well-defined key performance indicators (KPIs). Stakeholder engagement should be integrated throughout the process, providing a continuous feedback loop to ensure early validation of proposed ideas. Lastly, the BSP should consistently evaluate wCBDC against alternative solutions for addressing the identified pain points (see Figure 6).

Figure 6. High-Level Foundational Action Items



26. For the use case of settling tokenized government bonds, collaboration with key stakeholders, particularly the BTr, and a thorough cost-benefit analysis are critical before advancing to the PoC or prototype phase. The BSP should first clearly define the policy objectives for the use case and evaluate key risks and benefits.³² If feasibility and suitability assessments support moving forward, the BSP should formalize collaboration with stakeholders, particularly the BTr and participating FIs, to establish an operational framework and timeline for co-executing the project and delivering the PoC. Upon stakeholder agreement on the value propositions and design of the wCBDC, the project can advance to the prototype stage, incorporating real-world testing and refining the design and technology further (see Figure 7).

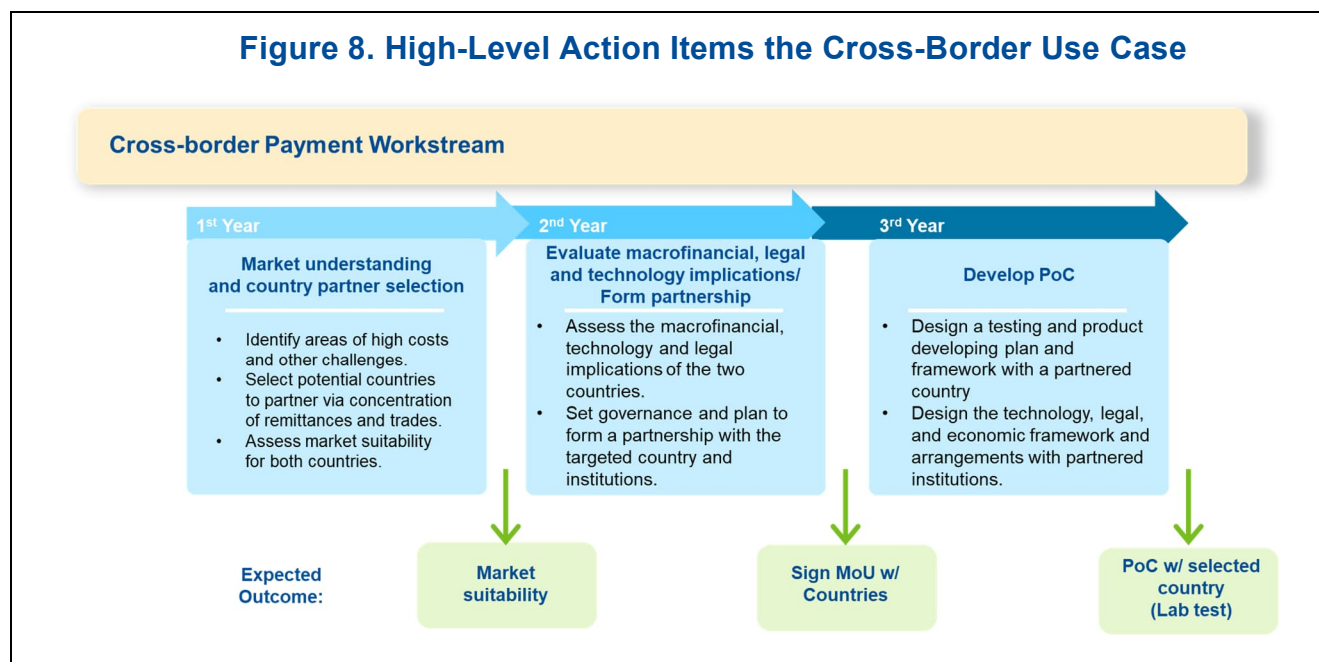
Figure 7. High-Level Action Items for the Tokenized Government Bond Use Case



27. Similarly, it is crucial to thoroughly assess the cross-border payment use case, including its benefits, risks, and policy implications for the Philippines and the prospective partner country before reaching agreement to proceed with the experimentation. The BSP should understand areas where wCBDC would be most suitable to improve cross-border payments, identify potential countries and understand the markets of both countries. Furthermore, both BSP and the prospective partner country should collaborate to assess key macro financial, legal, and technological implications from allowing wCBDC transactions between the two countries, as well as set up governance, resources, and timeline to execute the experimentation, if deemed desirable to pursue. After the formal agreement, the two countries should collectively plan the testing and select technology, vendors, and appropriate designs of wCBDC to conduct the PoC (see Figure 8).

³² For example, BSP could consider relate the policy objectives, benefits and risks on the potential use wCBDC for tokenized government bond settlement to support the following target outcomes for promotion of an inclusive, innovative and healthy financial sector under Chapter 11 (Ensure Macroeconomic Stability and Expand Inclusive and Innovative Finance) of [the Philippine Development Plan 2023-2028](#), as well as cross-cutting strategies outcomes related to fostering capital market development and promote safe and efficient National Payment System.

Figure 8. High-Level Action Items the Cross-Border Use Case



C. Recommendations

28. As the BSP continues to refine and advance this preliminary roadmap, the mission team offers several recommendations in different areas to ensure its successful implementation. As the BSP continues its wCBDC exploration, it is essential to integrate both policy considerations and technological developments at every stage. This holistic approach will enable the BSP to align its efforts with strategic objectives, effectively address potential challenges, and maximize the benefits of the wCBDC initiative.

Resource/ Project Management

29. Assign BSP staff with diverse, multidisciplinary expertise to the dedicated working teams for each use case. The working teams should include professionals not only with knowledge of payments systems but also with expertise in monetary policy, financial markets, technology, and legal and regulatory frameworks. This will ensure that critical steps of the use cases are adequately addressed by specialists who can contribute to informed decision-making. The working teams should leverage their combined expertise to tackle complex challenges, identify innovative solutions, and ensure alignment with the BSP's objectives.

30. Identify overlapping action items from the identified use cases in the roadmap and assess opportunities to optimize efforts and streamline process across both workstreams. This involves conducting a thorough review of tasks assigned for each use case to pinpoint areas of alignment and shared objectives. By consolidating resources and activities where appropriate, the BSP can reduce duplicating efforts and manage resources more effectively.

31. Examine the dependencies of the action items for each use case and allocate timeline for their completion. This involves identifying interrelated tasks and mapping their logical progression. By establishing clear dependencies, the BSP can anticipate bottlenecks, manage risks, and prioritize critical actions. Additionally, allocating a timeline for each action item is essential to maintain momentum and meet milestones. This includes consulting stakeholders to estimate realistic durations, considering resource availability, and accounting for external factors. Regularly reviewing and adjusting these dependencies will help the BSP remain agile and ensure smooth progress toward its roadmap goals.

Communication

32. Continue to engage and communicate with stakeholders to clarify the value propositions of wCBDC's identified use cases and gain their buy-in for the roadmap implementation. The BSP should actively engage and communicate with stakeholders, including those who missed the workshops, to clarify the value propositions of wCBDC use cases and align on potential benefits for the Philippines' payment systems. Targeted outreach initiatives, such as follow-up meetings and presentations, will share workshop insights, update stakeholders on the roadmap's progress, and address concerns. These engagements will also enable stakeholders to provide valuable input and evolving needs regarding digital money and payments.

33. Develop a joint communication strategy and outlets with key partners to ensure consistent messaging throughout various stages for implementation of each use case. Key partners may include BTr for the tokenized government bond use case and the central banks of partner countries for cross-border payments. The communication strategy should outline key messages, objectives, and target audiences for each phase, ensuring stakeholder alignment and incorporating feedback for future adjustments. It should present a unified narrative conveying the goals and benefits of the wCBDC initiative while addressing potential concerns. Additionally, it should identify suitable channels—such as press releases, webinars, and public consultations—to effectively engage various audiences, including FIs, policymakers, and the general public.

34. Promote public awareness and education to support digital and financial literacy. For the tokenized government bond use case, if financial access to government bonds would be identified as one of the key benefits of wCBDC, it is essential to accompany this initiative with robust public communication and education efforts to enhance digital and financial literacy among the public. The BSP and relevant organizations could actively promote awareness about the availability of the low-denominated, fractionalized government bonds.

Success Measurement and Continuation of the Roadmap

35. Continue to work within the 5P methodology and establish KPIs, go/no go decision points and success metrics for each wCBDC use case to measure progress and effectiveness. These indicators should include both quantitative and qualitative measures, such as transaction volumes, technical uptime, and customer satisfaction, aligning with the BSP's ongoing experimentation activities. The BSP should utilize the IMF's 5P methodology to set go/no go decision points to evaluate whether to proceed to a more advanced technical development or adjust initiatives based on KPI evaluation. This data-driven approach will facilitate agile responses to market conditions and support successful implementation of the wCBDC use cases.

36. Continue to refine the draft of the roadmap and engage with other internal departments and external stakeholders. The BSP should continuously refine the preliminary roadmap based on feedback from various sources. This iterative process will help clarify objectives, foster collaboration, effectively manage resources, and ensure timely achievements of the outcomes.

37. Ensure a proper legal framework to support experimentation and potential implementation. To effectively advance the development of wCBDC, it is crucial to establish a robust legal framework that outlines the regulatory landscape and operational guidelines governing its implementation. A legal framework for tokenized government bonds and cross-border wCBDC could also be considered and evaluated, noting the intention to experiment with wCBDC use cases. These frameworks should not only address compliance with existing financial regulations but also adapt to the innovative nature of wCBDC, tokenization, and cross-border payments, thereby providing legal clarity and certainty for all stakeholders involved.

Gather feedback and empirical data to assess and quantify the impact of government bond tokenization and wCBDC on the capital markets. The IMF may welcome potential follow-up missions or joint research projects to evaluate insights and data. The findings could help quantify the impact of tokenized bonds and wCBDC for the capital markets in the country.

Product-Market Fit and Suitability

38. Define clear policy goals and develop initial hypotheses that could be tested and evaluated for its economic impact. The BSP should clearly define its policy objectives for each use case, such as reducing transaction costs, promoting capital market development, and fostering financial innovation. Developing clear initial hypotheses is essential for evaluating these goals, such as reduction in settlement time and risks from using wCBDC. These hypotheses should undergo empirical analysis and the results should be assessed against alternative solutions in order to provide insights into the economic impact of wCBDC. This recommendation aligns with the earlier suggestion to establish success metrics for wCBDC experiments.

39. Engage stakeholders in the co-design of a wCBDC to ensure it delivers meaningful benefits to users and supports practical, real-world applications. Stakeholders can provide valuable insights into functional requirements, integration challenges, and emerging use cases. The BSP should facilitate co-design efforts with stakeholders that go beyond high-level consultation, by fostering more granular and iterative engagements such as structured feedback loops, targeted working groups, or user testing in realistic settings. This approach will help ensure the wCBDC addresses actual market needs, complements existing systems, and avoids unintended barriers to adoption.

40. Explore appropriate incentives and pricing structures to promote sustained adoption. The BSP could explore and implement appropriate incentives and pricing structures. FIs may be encouraged to adopt wCBDC through benefits such as reduced transaction fees, enhanced operational efficiencies, and access to new markets. By fostering an ecosystem where incentives align with the goals of all stakeholders, we can create a compelling case for the long-term adoption and engagement of wCBDC.

41. Continue researching technical and economic opportunities that tokenization could concretely bring to Philippine financial markets. As specified in the roadmap, it is essential to continue researching specific technical and economic opportunities that this innovative approach can offer to the Philippine financial markets. The BSP can further conduct in-depth analyses of existing market conditions and identify key areas where tokenization can provide tangible benefits; stakeholders can uncover opportunities for increased efficiency and transparency.

Technology

42. Collaboratively undertake research on technology solutions for wCBDC and tokenization experimentations. Given that potential partners involved in wCBDC use cases may hold varying perspectives and levels of understanding regarding these technologies, it is essential to collaboratively build capacity in this area. This involves not only gaining insights into the benefits and risks associated with different technological solutions but also fostering a shared knowledge base among all stakeholders. By doing so, the BSP can ensure alignment in selecting the appropriate technology and vendor for the experimentation of use cases, ultimately facilitating a more effective and cohesive implementation.

43. Continue researching and implementing measures to ensure appropriate management of related digital risks, such as cyber security and operational resilience. The BSP has already been closely considering digital risks and cyber security as part of its Project Agila work. It should continue conducting this research, noting the guidance provided by the mission team's desk review (see Annex V. Desk Review: Project Agila Outcome Reports).

44. Continue to improve upon technical shortcomings, deficiencies, and risks identified in the Project Agila’s sandbox. These include Project Agila’s reported challenges, such as maintaining data confidentiality and the high error rate in some scalability tests. They also include other pertinent areas of research and consideration, such as the technical ability for the BSP to intervene in transactions when errors or illegal activity occur, and the risks involved with providing access to external system administrators. In addition, the BSP could consider other technology choices such as permissioned DLT for its subsequent work.

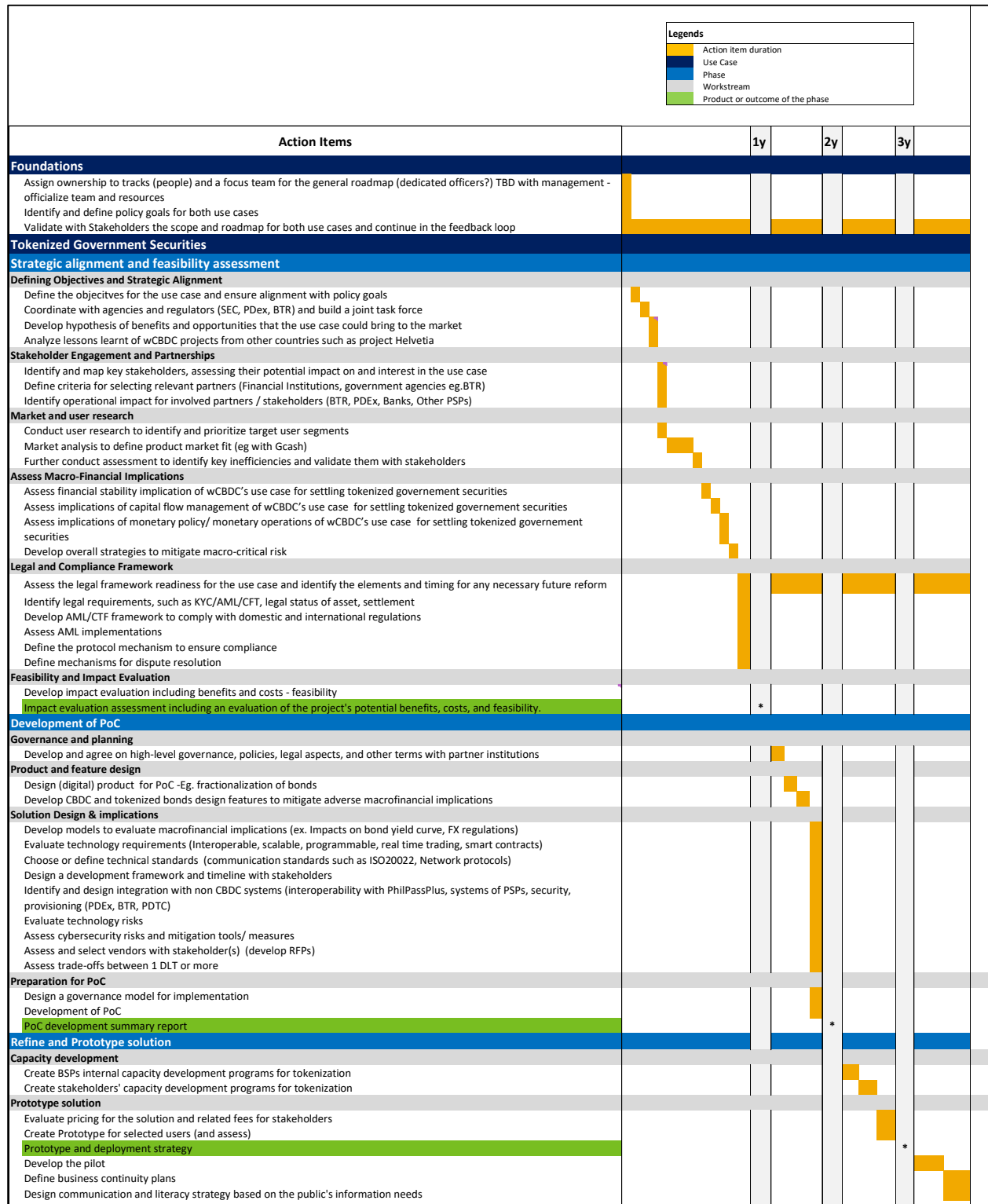
V. Conclusion

45. The mission facilitated stakeholder engagement and roadmap workshops, which played a crucial role in uncovering potential use cases for wCBDC and in developing a preliminary roadmap for the BSP. Among the identified use cases included in this preliminary roadmap were the application of wCBDC for settling tokenized government bonds and facilitating cross-border payments. These use cases emerged from valuable input provided by stakeholders, highlighting the need to improve individual investors' accessibility, promote secondary market trading, and address existing inefficiencies in cross-border payment processes.

46. The preliminary roadmap consists of action items, dependencies, durations, and allocated teams, which the BSP would need to further refine to move it forward. The roadmap is organized into three distinct workstreams. The first workstream focuses on critical foundational actions necessary for enabling wCBDC use cases, which includes formally establishing the project team and allocating essential resources. The second and third workstreams are dedicated to exploring the two previously mentioned use cases. Adhering to the 5P methodology, the first year will prioritize the definition of clear policy objectives, an analysis of costs and benefits in comparison to other alternatives, and the development of capacity both within the BSP and among potential external partners. This initial phase will lay the groundwork before later advancing to more complex exploration stages that will require a greater investment of time and resources.

47. As the BSP progresses with the preliminary roadmap, the mission provided recommendations in several aspects to ensure the successful implementation of the wCBDC initiatives. Effective communication is essential for engaging stakeholders to identify pain points and opportunities. A unified communication strategy will ensure consistent messaging while enhancing digital and financial literacy through targeted education. Resource management entails forming dedicated teams within the BSP to leverage multidisciplinary expertise and identify overlapping actions. Legal support is needed to establish a framework for the roadmap's action items. Additionally, a strong understanding of the technology stack is vital for adapting designs and managing digital risks. Research and collaboration with the stakeholders will ensure wCBDC's design and use cases are aligned with users' needs and the policy goals, while assessing wCBDC against alternatives. Finally, a success measurement will involve setting metrics and refining the roadmap based on the stakeholders' feedback.

Annex I. Preliminary Roadmap



Action Items		1y	2y	3y
Cross Border Payment				
Market understanding				
Market analysis and partner selection				
Identify and map key stakeholders				
Define criteria for selecting potential jurisdiction or country partners relevant remittances and trade Finance				
Refer to jurisdictions with relevant experiences				
Selecting the two countries to focus (one for remittances and one for trade finance)				
Assess macro-critical implications, and how to mitigate them consistently with target country goals				
High level market overview for the two countries		*		
Learning and Domestic Cooperation				
Analyze best practices and lessons learnt from other countries				
Identify and develop plan for cooperation with relevant domestic players and sign cooperations agreements				
Evaluate macrofinancial/ technology implications, and form partnership				
Financial and Economic Implications				
Assess implications of financial stability of wCBDC for the use case				
Implications for capital control				
Assess implications of capital flow management of wCBDC for the use case, also with target country's practices				
Assess implications of monetary policy/ monetary operations, of wCBDC for the use case				
Develop overall strategies to mitigate macro-critical risk				
Develop wCBDC design features to mitigate macro risks				
Develop models to evaluate impact on macro-metrics (ex. impacts on bond yield curve, FX regulations)				
Legal and Regulatory Compliance				
Assess the legal framework readiness for the use case and identify the elements and timing for any necessary future reform				
Assess AML implementations				
Develop AML/CTF framework to comply with domestic and international regulations				
Define the protocol mechanism to ensure compliance				
Identify legal requirements, such as KYC/AML/CFT, legal status of asset, settlement				
Define mechanisms for dispute resolution				
Check other target jurisdictions' legal frameworks to ensure they can support the use case experimentation or potential issuance of cross-border wCBDC				
Value Proposition and other solutions analysis				
Develop hypothesis of benefits and opportunities that the use case could bring to the Philippines' market				
Evaluate wCBDC's value propositions against other possible solutions				
Technical and operations considerations				
Evaluate BSP technology requirements				
Evaluate other target jurisdictions' technology requirements				
Governance and agreements				
Develop and agree on high-level governance, policies, legal aspects, and other terms with partner institutions				
MoU signed with the two relevant countries			*	
Development of PoC				
Testing and Development				
Design a testing and developing and prioritizing scheme (one country)				
Choose or define technical standards				
Benchmark different implementations for the same use case				
Solution Design & implications				
Design the technology, legal, and economic scheme of new arrangement with partner institution				
Evaluate pricing for the solution and related fees for all stakeholders				
Identify operational impact for involved partners/stakeholders				
Assess and selecting vendor(s) with Stakeholders				
PoC development summary report				*
Solution Design & implications				
Design a governance model for implementation				
Evaluate technology risks				
Assess Cybersecurity Risks and Mitigation Tools				
Assess if tech supports use cases (including if 1 DLT or more)				
Market analysis to define product market fit with stakeholders				
Conduct user research to identify and prioritize target user segments				
Assess integration with non-CBDC systems				
Further conduct assessment to identify key inefficiencies & validate it with stakeholders				
Explore innovations and business models for the evolution of the use case				
Internal and external Capacity Development				
Design awareness and literacy strategy based on the public's information needs				
Develop BSP internal capacity development programs				
Develop stakeholders and market capacity development programs				

Annex II. The Philippines's Wholesale Payment Landscape

The BSP has been working continuously to improve its wholesale payment system, in parallel to its wCBDC initiatives. Below is the summary the infrastructures relevant for wholesale payment system and their recent developments (Annex II. **Figure** provides the overview of the Philippines' national payment system, highlighting the wholesale system in red and the retail system in green).

- **The Philippine Payment and Settlement System Plus (PhilPaSS^{plus})** is the Philippines' real-time gross settlement system for processing and settling wholesale interbank payments, owned, operated, and overseen by the BSP. It also facilitates the settlement of fixed income security trades, foreign exchange trades, and other financial market transactions. Payments are conducted through Demand Deposit Accounts (DDA) maintained by participating institutions at the BSP. As of November 2024, there are 243 participants of PhilPaSS^{plus}, including banks, nonbank financial institutions, EMLs, sponsored participants, government agencies, BSP departments conducting monetary operations, FMLs, CSOs, and critical service providers (CSPs).³³

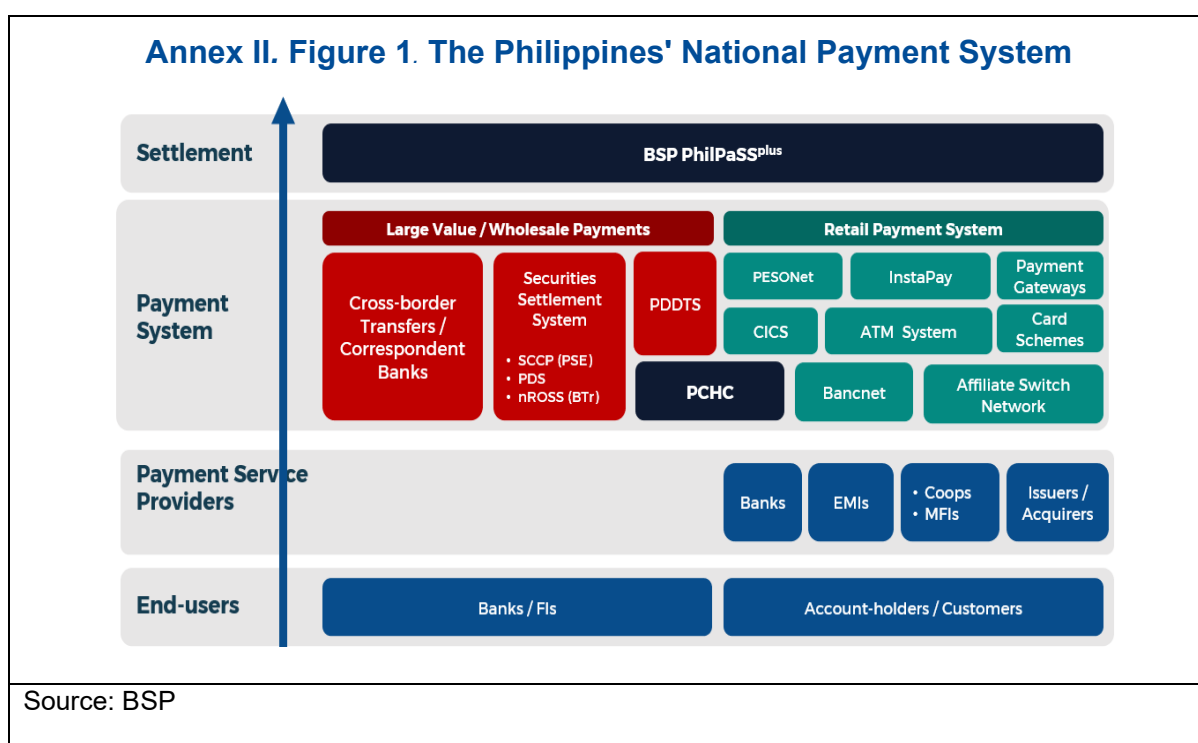
In 2021, PhilPaSS was revamped to PhilPaSS^{plus}. The key upgrades involved 1) *the adoption of ISO 20022 international messaging standards* (the migration is expected to be done by 2025), 2) *expanded standard operating procedure for holidays/work suspensions*, 3) *expansion of access* (for standalone EMLs, trust entities, cooperatives), 4) *improved business continuity measures*, and 5) *the intraday settlement facility (ISF)* for participants needing additional liquidity during daily operations.³⁴

- **The Philippine Clearing House Corporation (PCHC)** is a private entity incorporated in 1977 and owned by all member banks of the Bankers Association of the Philippines (BAP), which also provides several other critical payment-related services for the Philippine financial system. PCHC's key operations include:
 - *Check Image Clearing System (CICS)*, a system that uses digital images of checks for electronic transmission.
 - *Philippine electronic funds transfer (EFT) operations network (PESONet)*, an electronic fund transfer service that allows batch processing of peso-denominated payments across banks.
 - *Payment Application Secure (PAS)*, a system that facilitates the collection of customs duties and taxes, and transmittance to the Treasury.
 - *The PCHC Payment Gateway (Paygate)*, which provides a means of communication of transaction information for effecting payments via PESONet.
 - *Philippine Domestic Dollar Transfer System (PDDTS)*, a system to facilitate the efficient transfer of USD funds within the Philippines.
- **The Philippine Dealing System (PDS) Group** operates a number of important infrastructures that facilitate trading, clearing, settlement, and post-trade functions. The PDS Group consists of a holding company, the Philippine Dealing System Holding Corporation, and several subsidiaries. There are many resources shared between these entities and the Philippine Stock Exchange (PSE), which is a major shareholder of PDS. For instance, compliance and monitoring are done by one single team for all subsidiaries. These subsidiaries include:
 - *Philippine Dealing & Exchange Corporation (PDEX)* provides trading infrastructure for the fixed-income market.

³³ CSPs are technology companies that provide solutions which are vital to the operation of PhilPaSS^{plus}.

³⁴ Source: BSP staff and [Peso RTGS Payment System Participants](#).

- *Philippine Depository & Trust Corporation (PDTC)* acts as depository, registry, and/or intermediary of participants for equities, commercial papers, and corporate bonds.
- *Philippine Securities Settlement Corporation (PSSC)*, the payment and transfer services arm and operates the real-time domestic transfer system for Chinese Yuan, through the Renminbi Transfer System (RTS).
- *PDS Academy for Market Development Corporation (PDSA)* provides market education and development.
- **Securities Clearing Corporation of the Philippines (SCCP)**, a Central Counterparty (CCP) operating as a central security clearing institution since 2002, matches all broker-to-broker equities transactions executed at PSE. The settlement of trades of listed securities takes place two business days after the transaction date or on T+2 on a delivery-versus-payment multi-lateral net settlement basis. SCCP is a wholly owned subsidiary of the PSE and regulated by Securities and Exchange Commission (SEC).
- **National Registry of Scripless Securities (nRoSS)** functions as a Central Securities Depository (CSD) and provides centralized electronic registration, custody, sole legal authority for the transfer of ownership, and management of scripless treasury bills and government bonds. It is operated by the Bureau of the Treasury (BTr), a public entity under the Department of Finance. nRoSS directly connects to PhilPass^{plus}, enabling the settlement of government securities via central bank money under a Delivery-versus-Payment (DVP) basis.



Annex III. Ongoing wCBDC Experimenting Projects and Use Cases

This Annex shares information and lessons from wCBDC experimentation around the world that relates to tokenized bond transactions and cross-border payments.

Tokenized Securities Payment and Settlement

Several projects have investigated how CBDC can be used to support tokenized securities settlement. Most projects are focused on how tokenized securities and wCBDC on the same platform can facilitate better DvP settlement. Generally, the experiments demonstrate positive results for *technical feasibility*. However, less focus is placed on the overall merits compared with other options, including upgrading existing systems, and technical and cyber resilience. Many experiments also assume (and thus seek to be prepared for) an environment where DLT-based and tokenized financial transactions will be in high demand in the market; however, the ultimate size and sustainability of tokenized securities markets are yet to be clear.

The Swiss National Bank (SNB) is one of the leading central banks to observe and learn from in this space. Project Helvetia is a collaborative experiment conducted by the BIS Innovation Hub (BISIH) Swiss Centre, the Swiss National Bank, and SIX Digital Exchange (a licensed Swiss financial infrastructure operator). The project aims to explore the settlement of tokenized assets in central bank money, focusing on the practicality and legal feasibility of using a wCBDC for such settlements. The project has undergone multiple phases: Phase I employed simulated tokenized assets and transactions to demonstrate the functional feasibility and legal robustness of settling tokenized assets with wCBDC. Phase II further explored these areas, showing that a wCBDC can be integrated with existing banking systems and an RTGS environment. Phase III is piloting real CHF wCBDC without the involvement of the BISIH. The SNB issues the wCBDC, for use only on an intra-day basis, on the privately operated SIX Digital Exchange. It is used for payment for primary market tokenized bond issuance on the same platform. The SNB is monitoring the pilot, including market appetite and operational soundness, for the next two years before making further decisions about potential wCBDC issuance.³⁵

Project Agila Could Learn from Insights Gained Through Project Helvetia's Multiple Phases:

- *Integration with Existing Systems:* Project Helvetia's experience in integrating wCBDC with existing core banking systems could offer valuable insights for the smooth integration of a potential wCBDC with the Philippines's financial infrastructure.
- *Legal and Functional Feasibility:* The exploration of legal frameworks and functional feasibility in Project Helvetia could serve as an example for the Philippines. Of course, a thorough assessment of legal implications in the specific context of the Philippines would be needed.
- *PoC and Pilot Testing:* The Philippines can draw lessons from the structured and phased approach of Project Helvetia, as another methodology for moving through PoC and pilot evaluation.

Other prominent and interesting experiments to learn from include:

The Reserve Bank of Australia (RBA): The RBA's most recent wholesale CBDC work is part of its Project Acacia.³⁶ It completed phase 1 of work in 2024, where it conducted technical tests involving wholesale CBDC as payment for tokenized asset transactions. It found benefits from initial study, in terms of cost and risk reductions when the tokenized money and assets were on the same programmable platform. Namely, the atomic exchange of money and assets on the same ledger enabled instantaneous,

³⁵ Source: [Project Helvetia report](#).

³⁶ Source: [RBA's Project Acacia phase 1 report](#).

simultaneous transfers, and programmability was found in theory to help improve efficiency with coupon payments on bonds. Phase 2 will be conducted in 2025 and involve testing prototypes with industry partners. The RBA writes, “A key question is whether there is a need for a new form of tokenised central bank money in the form of a wholesale CBDC to maximise the benefits from asset tokenisation, or whether there are enhancements to existing infrastructure that could support settlement using ESA [RTGS] balances, perhaps complemented by privately issued tokenised money such as tokenised deposits” (p. 21).³⁷

South African Reserve Bank (SARB): Project Khokha phase 2 notes that the use of a single shared ledger with different levels of access, depending on the role of the participant, enabled greater oversight of what was happening across the network, making reconciliations easier and providing valuable real-time operational data. Having both the security and settlement tokens on DLT enabled DvP on a single network; however, the porting of the central bank’s settlement token across two platforms also introduced some challenges, for instance, in the separation between operational and legal settlement introduced by porting the wCBDC token to the Khokha Hub. The project concluded that issuing a token-based wCBDC on DLT created a riskless settlement asset on a platform that also serves as a payment settlement system and hosts settlement participants’ accounts as a wallet on the DLT with the central bank.

Participants have observed that the true benefit and challenge from DLT may not stem from the technology itself, but from the business process innovation it enables. Based on the PoC, although the number of operators could potentially be reduced, the roles and functions did not disappear and therefore existing regulatory frameworks would still apply. The design of the PoC did, however, result in the issuance of DLT-based tokens, which are not defined in law, and the modular design of platforms would require further consideration to determine whether and how frameworks may need to be updated.³⁸

Bank of Canada (BoC): Phase 3 of Project Jasper explored how payments and settlements could work within a distributed ledger by developing a prototype system integrated with existing market infrastructures. This phase introduced securities and cash onto the ledger via Digital Depository Receipts (DDRs) created by the CDS and the BoC, allowing participants to execute securities settlements using central bank funds.

The experiment revealed promising aspects of the DLT platform, such as effective integration with financial market infrastructures through a “loose coupling” approach, enforcement of independent token issuer rules, and participation restrictions via a private ledger and doorman service. It also provided real-time ledger transparency for market infrastructure and regulatory observers. Additionally, the PoC established a credit extension mechanism for brokers outside the payment system, allowing access to cash DDRs—reflecting existing credit provisions. However, this reintroduced credit risk in a model aiming to eliminate it.

The project indicates that DLT models implementing significant changes to current settlement systems could yield operational cost savings and reduce reconciliation efforts. Ultimately, while DLT shows potential for efficiency improvements, expanding the ledger’s scope to include more assets and the entire trade lifecycle may be essential for realizing these benefits.³⁹

Banque de France (BdF) has also launched an ambitious wCBDC program including an initiative where the BdF worked with the European Investment Bank (EIB) and Société Générale – forge to settle a European Investment Bank (EIB)-issued digital bond with wCBDC.⁴⁰

Project Agila could study the reports and work of these efforts to advance its understanding. The projects note some important caveats. First, they are largely exploratory studies where the central banks have not yet made decisions about whether to issue wCBDC and if it would be the most suitable

³⁷ Ibid

³⁸ See [SARB 2022](#) for more about Project Khokha.

³⁹ See [BoC 2022](#) for more on Jasper Phase 3.

⁴⁰ See [BdF 2023](#) for more on their experimentation.

intervention for improving wholesale markets. Also, they note that addressing regulatory and legal challenges from the outset is critical for compliance and success.

Cross-Border Wholesale CBDC Projects

Exploring wCBDCs in cross-border transactions has seen new experimentation that focuses on addressing the challenges and identifying the benefits of such digital currencies in international finance. Key experiments include:

Project Jura, a collaboration by the Bank for International Settlements (BIS), Banque de France (BdF), and Swiss National Bank (SNB), focused on the cross-border settlement of wholesale Central Bank Digital Currencies (wCBDCs). Jura tested the feasibility of using DLT to secure and efficiently transact two distinct wCBDCs. Key innovations included the deployment of digital currencies on third-party platforms and the introduction of mechanisms like subnetworks and dual-notary signing to facilitate secure and policy-compliant transactions.⁴¹

Project Jasper-Ubin was a cross-border collaboration between the Bank of Canada and the Monetary Authority of Singapore, designed to explore the use of blockchain technology for the clearance and settlement of payments and securities. Utilizing DLT, the project aimed to improve the efficiency of cross-border transactions by enabling direct currency exchanges without the need for intermediaries. The experiment successfully demonstrated how two different blockchain platforms could be linked to facilitate real-time, cost-effective, and secure cross-border payments, thereby showcasing the potential of DLT in enhancing global financial transactions.⁴²

Project Dunbar was an international initiative led by the BIS Innovation Hub, collaborating with the Reserve Bank of Australia, Bank Negara Malaysia, the Monetary Authority of Singapore, and the South African Reserve Bank. This project aimed to develop prototypes for shared platforms that could enable banks in different countries to transact directly with each other using digital currencies issued by multiple central banks. The key achievement of Project Dunbar was the demonstration of the feasibility of a common platform for cross-border transactions using multiple CBDCs, which could significantly reduce reliance on intermediaries and improve the efficiency and security of international payments.⁴³

Project mBridge is a venture led by the BIS Innovation Hub, which collaborates with the central banks of China, Hong Kong, Thailand, and the United Arab Emirates.²² The project's goal is to create a cross-border corridor network that facilitates more efficient international trade and investment through wCBDCs. By leveraging DLT, mBridge aims to simplify the process, reduce costs, and increase the speed of cross-border payments and currency exchanges. The project has successfully demonstrated the potential of CBDCs to transform the landscape of international finance by enabling faster, cheaper, and more transparent transactions. Project mBridge continues its development and has reached the minimum viable product (MVP) stage, while broadening its international reach. In October 2024, the BIS announced the project will graduate from the list of BIS IH projects.⁴⁴

⁴¹ See <https://www.bis.org/about/bisih/topics/cbdc/jura.htm> for more on Project Jura.

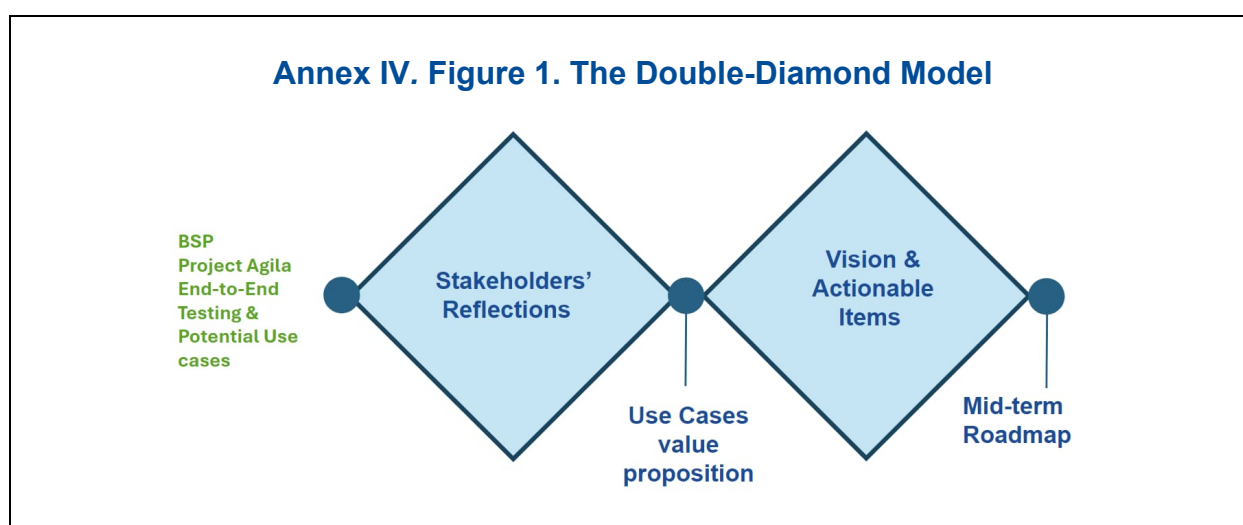
⁴² See <https://www.mas.gov.sg/-/media/Jasper-Ubin-Design-Paper.pdf> for more on Jasper Ubin.

⁴³ See <https://www.bis.org/about/bisih/topics/cbdc/dunbar.htm> for more on Project Dunbar.

⁴⁴ See https://www.bis.org/about/bisih/topics/cbdc/mcbdc_bridge.htm for more on Project mBridge.

Annex IV. Description of Methodologies Used During the Workshops

Following a Human-Centered Design approach, the mission team applied the Double-Diamond model as the foundational framework for the design process, customizing it to suit the mission's unique requirements of identifying the value for each use case and developing a preliminary roadmap (see Annex IV. **Figure**). This model facilitated the mission team in guiding participants through a process of divergent and convergent thinking, enabling a detailed examination of the barriers and opportunities within the Philippines' payment ecosystem and clustering them into action items on a roadmap. Ultimately, it allowed the team to pinpoint persistent gaps where wCBDC could make a significant impact and prioritize use cases deserving of further exploration. Table 3. below summarizes how the process was implemented during the mission.



Annex IV. Table 1. Workshop Plan and Objectives

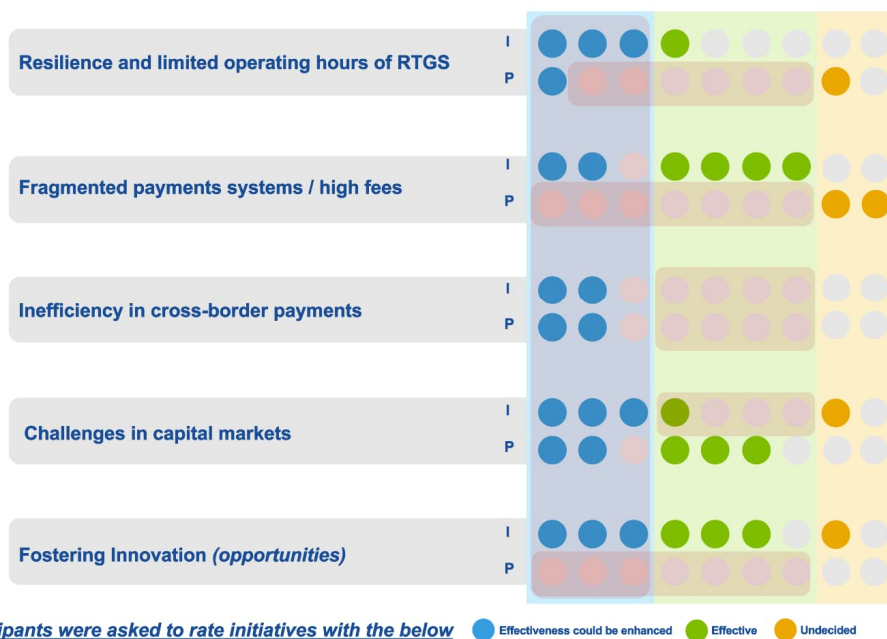
Workshop details	Goal of the workshop	Method and dynamics
Day 1		
Pain Points validation Duration: 95 minutes	Engage with stakeholders to identify and validate current challenges and opportunities, ensuring a comprehensive understanding of the issues and their broader implications. Key questions to address: What are the primary pain points within the Wholesale Payments Landscape? How do these challenges specifically impact your type of institution?	In a group setting, the mission team presented the key challenges and opportunities identified by participants during the development of Project Agila. Participants were then invited to individually reflect on the critical issues facing the Philippines' wholesale payments ecosystem, considering their practical implications for businesses and customers. Facilitators subsequently guided a discussion where participants shared insights and explored the underlying causes of these challenges.
Current and Planned Initiatives	Building on the pain points identified in the previous activity, participants were	After the group discussion, participants individually mapped existing initiatives or

Duration: 1.45 hours	asked to pinpoint relevant market-level initiatives that have been implemented, are planned, or could be proposed to address these challenges. They also assessed and ranked these initiatives based on their effectiveness in resolving the identified pain points. Key questions to address: What current or planned initiatives are you aware of that address each of the pain points? What new initiatives could be introduced to help mitigate these challenges? How effective do you consider the current and planned initiatives to be?	solutions currently being implemented to address the identified challenges. The group then collectively evaluated the effectiveness of these initiatives, classifying them into three categories: "effective," "could be enhanced," or "unclear." This exercise enabled the mission team to identify potential gaps and areas where CBDC could play a significant role (see Annex IV. Figure 2).
Day 2		
Value Proposition and Prioritization of Use Cases Duration: 2.5 hours	Evaluate the pros and cons of each use case, identify relevant stakeholders, and outline key unanswered questions to comprehensively map the current understanding of each use case. This process aims to highlight limitations and pinpoint areas requiring further exploration, providing essential inputs for the development of the subsequent roadmap. Key questions to address: Based on the main challenges identified, what potential use cases should be considered for the Philippines? What are the opportunities and challenges associated with each use case from your perspective? Who should be involved in developing these use cases, and how can the BSP effectively collaborate with industry and other stakeholders?	At the start of the session, the IMF presented a summary of key insights from the Day 1 workshop. To provide stakeholders with a foundational understanding of wholesale CBDC, the mission delivered presentations on "Wholesale CBDC 101" and explored emerging trends in digital payments and technology, including the opportunities presented by tokenization and programmability. Following this, participants engaged in a group discussion to analyze the prioritized use cases from Day 1. They explored opportunities, risks, stakeholders, and unresolved questions, completing a value proposition for each use case. Afterward, participants voted on the use cases they deemed most impactful for the payments landscape and most feasible within the current ecosystem. (Annex IV. Figure shows the results after participant voting on Day 2)
Roadmap workshop (IMF & BSP)		
Vision Framing & Road Mapping	Develop a medium-term vision for the evolution of each use case, accompanied by a preliminary set of actions on a roadmap. This visioning exercise employed a "Future Thinking" approach, working	To develop vision statements, participants were divided into two groups, each focusing on a specific CBDC use case. The first group envisioned a newspaper cover story from 2029 centered on cross-border transactions, while the second group crafted a narrative for the

	backward from the desired future state to the present. This method ensures a clear alignment of activities with the formulated vision, creating a structured path informed by the value proposition of each use case. Key question to address: For each specific use case, what are the key success factors or transformative outcomes from the BSP's perspective that could serve as game changers?	tokenization of government securities. Both groups were tasked with creating an engaging title and identifying the key highlights of their envisioned success stories. With these future visions as a foundation, the entire team collaboratively outlined the milestones to be achieved in the medium term (three years). Subsequently, each group was provided with a set of preliminary action items derived from the value propositions identified by stakeholders. Participants were encouraged to customize their roadmaps by adding, removing, or refining action items to incorporate case-specific details as needed.
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Annex IV. Figure 2. Solution Gaps for the Identified Challenges

Effectiveness impression from participants on *planned (P)* and *in-place (I)* initiatives



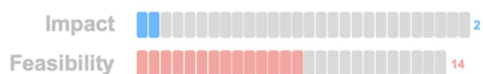
Annex IV. Table 2. wCBDC's Potential Use Cases and Value Propositions

wCBDC use cases	Key Benefits / Opportunities	Key Risks / Concerns	Other considerations	Stakeholders / Potential collaboration
wCBDC for cross-border payments	Open more access for people to invest in global securities, ease of payments for foreign players, level playing field, support for USD independence, 24/7 availability, standardized messages/data fields, more channels for receiving investors, real-time tracking of investments, minimize reconciliation efforts, fractional ownership, 24/7 settlements, better security and efficiency, potential for more efficient secondary market, programmability and smart contracts, support for retail investors, lower costs, improved reliability.	Cyber risks, money laundering, operational risks, profitability in banks, content inconsistency, legal concerns, systemic risk from settlement failures, PHP restrictions as a currency, drawing down on reserves, cost of maintaining systems, limited demand based on country size, lack of expertise, transparency and traceability, intra-Asian trade finance dependencies.	Whose benefit does it serve, cost absorption, security assurance, AML safety, standardized regulations, partner prioritization, convincing others to engage, data privacy and AML control, exchange rate mechanisms, bilateral/multilateral considerations, cost accountability, latency improvements.	Regulators, PSPs, FIs, tech vendors, sandbox collaboration, BIS, SWIFT, local payment management boards, customs, other central banks (CBs), G20, related global bodies.
wCBDC for settling tokenized government bonds	Real-time tracking of investments, minimize reconciliation efforts, fractional ownership, 24/7 settlements, better security and efficiency, potential for more efficient secondary markets, programmability and smart contracts, support for retail investors, lower costs, improved reliability.	How secure is the protection on the ledger? Participant inclusion, liquidity, BCP, financial literacy, policies for issuance, design criteria for HQLA, suitability for retail investors, potential costs.	Will it drive costs down, is it lucrative versus bond funds, maturity mismatch, dispute management, platform selection, availability to retail investors, secondary market trading, gas fees, number of players.	BSP, BTR, PDTC, FIs, SEC, DOF, clearing operators, PPMI, IMF, central banks, regulators.
Additional RTGS functionalities on wCBDC	24/7 payments, additional investment venues, real-time monitoring of positions, more payment system options, less volatile exchange rates, extend business opportunities, supports innovation, serve more customers, acts as a similar facility, eliminates cut-offs, operational readiness for CBDC, less reconciliation, dedicated venue.	Operational risks, fungibility issues, setup costs, industry support, impact on banking systems, new costs for maintaining CBDC, technology risks, regulatory and market risks, interoperability costs, business need justification, cost vs. traditional systems.	Holistic design structure, CBDC requirements, technical requirements, interoperability costs, accountability, how to determine exchange rates, access policy requirements, feasibility as an RTGS alternative, auditing requirements, transaction finality, DDA conversion, data recovery strategies, transaction types allowed.	BSP, PPMI, industry associations, FIs tokenizers, CSD, PCHC, technology vendors, working groups, committees, payment system operators, central bank stakeholders, industry players, technology providers.

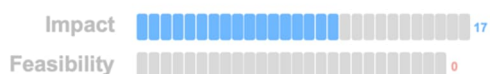
Annex IV. Figure 3. Impact and Feasibility Results After Participant Voting

Feasibility / Impact

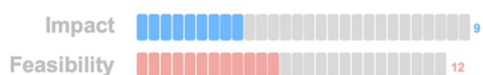
1 Test more functionalities of RTGS on wCBDC



2 Cross-border CBDC

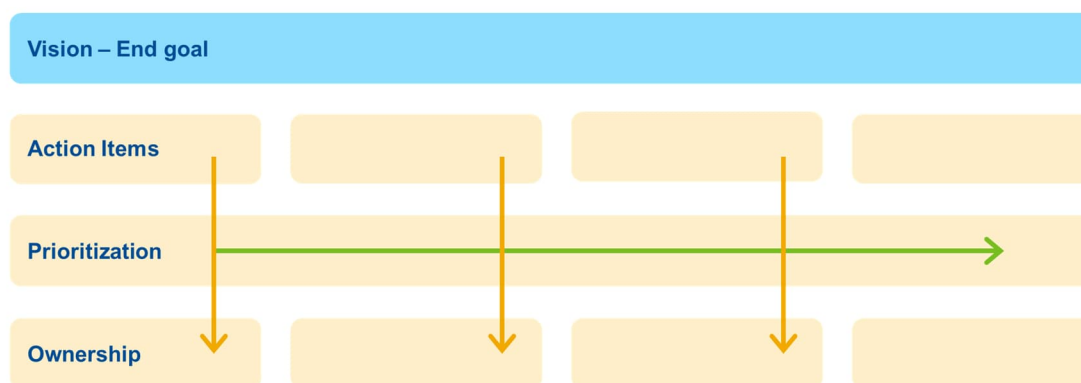


3 Support tokenized government securities settlement



Annex IV. Figure 4. Roadmap Exercise

Activity Flow



Annex IV. Figure 5. Vision Framing

Cross Border

- Emerging digital payment **leader** for cross border transactions in **Asia**.
- **Cross border** digital payment conqueror.
- PH adopts wholesale CBDC: **quicker, safer and more cost-effective settlement**.
- BSP successfully issues CBDC for cross border payments that support **trade and remittances**.
- PH unlocked the power of CBDC in facilitating remittances for Filipinos **across the world**.

Tokenized Bonds

- PH leads the establishment of tokenized bonds **operational framework with industry partners** / players.
- Bonds are made affordable for **retail investors**.
- Adopt tokenized assets to streamline **securities trading**.
- Pushes **financial inclusion further** through more accessible investments.
- PH financial system undergoes change towards more **inclusive, sustainable access** to investors.

Vision 3 years from now

“BSP unlocks the power of CBDC, pushing the Philippines to become a leader in Asia for cross border payments, creating a quicker, safer, and more effective settlement for trade and remittances across the world.”

“Philippines’ financial system undergoes transformative change by setting an operational framework with industry partners, pushing financial inclusion through more inclusive access to investors.”

Annex V. Desk Review: Project Agila Outcome Reports

The IMF mission team provided feedback to the Project Agila preliminary outcome report. This feedback constituted a desk review that the mission team conducted in parallel to the onsite mission. In this desk review, the IT Dept. and MCM reviewed the “Report on Project Agila Results, Technical Findings and Policy Implications” and supplemental annexes that pertained to technology aspects of the project. A brief summary of the desk review is below.

The IMF mission team commends the BSP for the thoroughness and clarity of the Project Agila draft report. The BSP has taken care to identify and document the prototype’s technology design, as well as number of technological challenges and shortcomings. The project report and annexes, including key technology issues and performance metrics, are so detailed that the mission team recommends the BSP consider publishing them to help guide other central banks.

A sample of the mission team’s technical recommendations for the BSP are summarized below, with more details in the full submission:

Governance and transaction validation process using the permissioned DLT:

- Node operating hours: Consider and clarify whether BSP node operates continuously or if banks solely handle validation during off-hours, and the risks that could arise from this.
- Clarify the required number or percentage of validating nodes for transaction approval, and the trade-offs associated with a greater or lower number.
- Consider the technical possibility for BSP to intervene to correct transactions when required.
- Review risks related to granting the private financial institutions system administrator rights.
- Clarify which financial institutions can access certificate authority nodes and assess related risks.

Smart contract risk management: Evaluate risks from allowing financial institutions to modify smart contracts on the wCBDC platform. Also, consider how to protect the code from unauthorized changes.

Technical challenges: Address technical issues identified in the annexes (especially annexes G and E), including challenges with data confidentiality and the 89% error rate in scalability tests.

Cybersecurity considerations, in addition to the report’s existing careful discussion:

- Define roles among cloud service provider, solution provider, and the BSP.
- Enhance user credential protections via access identity management (AIM).
- Formulate policies to manage risks from third-party vendors and suppliers.
- Implement and test incident response and recovery plans.

The BSP PMT is also advised to explore the following resources: (1) [BIS Report on DLT Digital Risks](#), (2) [WEF Paper on CBDC Technology Considerations](#), (3) [Project Leap on Quantum Computing](#)

The discussion of policy and macro-financial implications is also carefully written. That said, some recommendations include the following:

- Opportunities to consider complexities around classifying CBDC and imposing holding limits (such as “waterfall” schemes and the timing of accounting limits in the 24/7 setting).
- Softening the language around the potential benefits of tokenization and wCBDC on financial markets (pages 34-37) and on financial inclusion (page 37) to reflect uncertainties and the need for country-specific analysis that accounts for the specific pain points in the market.

Annex VI. BSP- IMF wCBDC Medium-Term Capacity Development Workplan

Actions/ activities (MCM and BSP)	Prerequisites ⁴⁵	Timeline ⁴⁶	Status
1. IMF (MCM CO and PI, LEG FFL and FIG, ITD) TA mission (remote) on wCBDC prototype: training, documents review, preliminary assessment.	BSP shared the data and materials necessary for a productive IMF TA mission and agreed on the project plan of activities and milestones.	July 15, 2022, to February 28, 2023	Completed
2. MCM (CO and PI) TA mission: (in-person) on wCBDC prototype; (in-person) wCBDC use-case analysis.	BSP followed the previous TA recommendation in its development of use-cases as well shared the data and materials necessary for a productive IMF TA mission.	December 1-14, 2022	Completed
3. Follow-up IMF (MCM CO and PI; LEG FFL and FIG; and ITD) TA mission: (in-person) to take stock of the BSP implementation and agree on the updates in this workplan.	BSP staff shared the data and materials necessary for a productive IMF TA mission and agrees on the main objective of the mission to update this workplan.	November 20-23, 2023	Completed
4. Post-mission engagements (remote) on documents for wCBDC prototype	- BSP staff shared the Project Agila systems requirement document and the sandbox operating guideline with the IMF team, which the team provided comments. - BSP shared the draft test plan for Project Agila that the IMF team also provided comments	January to March 2024	Completed
5. BSP completes sandbox test of a wCBDC prototype	BSP follows the previous TA recommendations.	By September, 2024	Pending the development of prototype
6. BSP shares the draft of Agila's final assessment report and IMF provides desk review (virtual)	- BSP follows the previous TA recommendations. - BSP shares the testing results of the project.	Mid-October 2024	Pending the development of prototype and the report
7. BSP completes the final assessment report	BSP follows the previous TA recommendations.	By January 2025	Pending the completion of the report

⁴⁵ These are expected to be in place in time for the respective activity to start as well as to add value as intended.

⁴⁶ The dates specified for future events are indicative only and may be subject to revision.

Actions/ activities (MCM and BSP)	Prerequisites ⁴⁵	Timeline ⁴⁶	Status
8. Follow-up IMF (led by MCM PI with ITD, and CSF) TA mission on stakeholder engagement (onsite): - Based on the final assessment of Project Agila, the mission team assists BSP in developing a stakeholder engagement plan and assess potential use cases of wCBDC, with a focus on design implications and associated risks. - The mission team will provide an internal seminar to cover key issues raised by the BSP such as macro-financial implications, tokenization, and cross-border aspects of wCBDC. - With CSF's support, the team plans to adopt design thinking techniques to assist the research process. - ITD will be tasked with the plan related to project management and technology issues. - The outcome of the mission will help support the BSP in crafting the wCBDC roadmap, aimed to publish in January 2025.	- BSP follows the previous TA recommendations. - BSP staff share the data and materials necessary for a productive IMF TA mission and agree on the main objective of the mission to update this workplan.	November 25 – December 3, 2024	Pending the publication of the outcome report
9. BSP to draft CBDC Roadmap, which will include details on strategic initiatives that will be pursued by the BSP in collaboration with payment system stakeholders.	BSP follows the previous TA recommendations.	By January, 2025	Planned
10. BSP to decide whether to issue a wCBDC.	BSP follows the previous TA recommendations (specific actions the BSP is expected to take).	TBD	TBD ⁴⁷

⁴⁷ This decision will mainly be anchored on the following factors, among others:

1. The outcome of the ongoing project;
2. Feedback of the payments industry based on their learnings in the project;
3. Specific wCBDC use cases which shall be jointly identified by BSP and payments industry as potential mitigants to address pain points in the national payment system;
4. Strategic priorities of the BSP; and
5. Advice from the IMF, as the BSP's wCBDC Project External Adviser.